

Linnet

Half-edge graphs and subgraph-first algorithms

1 Overview

Linnet is a graph library for tensor networks, Feynman diagrams, and other workflows where the boundary of a subgraph matters. Its central representation is a half-edge graph. An ordinary internal edge is a pair of half-edges, while an external edge has one dangling half-edge. A node owns incident half-edges, and a subgraph is a selection of half-edges rather than a detached copy of the graph.

Why half-edges

Cutting an internal edge can turn its two halves into boundary edges without changing the degree of either endpoint. This makes operations such as extraction, excision, contraction, sewing, traversal, and cut analysis natural for physics graphs. Linnet algorithms therefore accept subgraph views even when the caller wants to operate on the full graph.

1.1 Choose a task

- To construct and validate a graph with a boundary, follow the [first-graph tutorial](#) and the exact `HedgeGraphBuilder` reference.
- To enumerate cycles, cuts, or connected regions, use the [graph-algorithms guide](#) with the `HedgeGraph` reference.
- To render a tree of existing DOT files and rebuild only changed figures, use the [Clinnet command-line guide](#).
- To lay out or render a graph, continue to the [Linnet Typst guide](#) layout coordinates are deliberately separate from graph identity.

1.2 A working model

A Linnet workflow has four parts:

- construct a graph with `HedgeGraphBuilder`, or parse a DOT graph;
- retain the typed identifiers returned for half-edges, nodes, and edges;
- describe the region of interest with one of the subgraph representations;
- run traversal or mutation operations against that subgraph and the graph's involution.

The involution is the pairing map for half-edges. A paired entry represents an internal edge; an identity entry represents a dangling edge and records its flow. Edge orientation and flow are related concepts, but they are not interchangeable. Consult the API reference for the exact variants and conversion rules in your Linnet release.

Indexes belong to one graph

`Hedge`, `NodeIndex`, and `EdgeIndex` are compact typed indexes, not durable object IDs. Graph edits can change storage and mappings. Keep the result returned by an operation, and do not apply an index or subgraph bitset to an unrelated graph just because its numeric values happen to fit.

1.3 Construction and interchange

The Rust package can be added directly from [crates.io](#):

```
[dependencies]
linnet = "0.17"
```

Programmatic construction uses `HedgeGraphBuilder`. DOT support provides a convenient human-readable interchange and debugging format; it can carry global, node, edge, and half-edge attributes. Treat DOT as a representation boundary, not as a substitute for the typed graph invariants. Validate or handle parser errors before running algorithms.

The drawing feature adds layout support. Layout is separate from graph identity: coordinates may change without changing the topology. Linnet itself does not provide a supported renderer; use `DOT`, `Clinnet`, or `Linnest` at the presentation boundary.

1.4 Related crates

One graph layer, several consumers

`Spensio` uses Linnet for tensor-network connectivity and adds tensor contraction semantics. `GammaLoop` uses the graph model in collider workflows and adds process generation, integration, and persistent state.

The rendered guides and generated APIs are the canonical usage reference. Contributors can follow the stores, invariants, mutation flow, and feature boundaries in the [Linnet implementation architecture](#).

2 Tutorial

In this tutorial, you will construct a half-edge graph in Rust, validate its invariants, find its one independent cycle inside an explicitly selected subgraph, and print DOT. The builder returns typed node indexes; Linnet's subgraph filters then select the half-edges on which an algorithm operates.

2.1 Prerequisites

Create a Rust binary project and add the current Linnet release:

```
cargo new linnet-first-graph
cd linnet-first-graph
cargo add linnet@0.17
```

No drawing feature or external layout program is needed to construct the graph and emit DOT.

2.2 Build a graph with a boundary

Replace `src/main.rs` with the following program:

```
use linnet::half_edge::{
    builder::HedgeGraphBuilder,
    involution::Flow,
    subgraph::{SubSetOps, SuBitGraph},
    HedgeGraph,
};

fn main() {
    let mut builder = HedgeGraphBuilder::new();
    let a = builder.add_node("a");
    let b = builder.add_node("b");
    let c = builder.add_node("c");

    builder.add_edge(a, b, "ab", false);
```

```

builder.add_edge(b, c, "bc", false);
builder.add_edge(c, a, "ca", false);
builder.add_external_edge(a, "incoming", false, Flow::Sink);
builder.add_external_edge(c, "outgoing", false, Flow::Source);

let graph: HedgeGraph<&str, &str> = builder.build();
graph.check().expect("the half-edge involution is valid");

let boundary: SuBitGraph = graph.external_filter();
let internal = graph.full_filter().subtract(&boundary);
let (basis, _) = graph.cycle_basis_of(&internal);
assert_eq!(basis.len(), 1);
println!("{}", graph.base_dot());
}

```

Run it with `cargo run`. Success means the assertion passes and stdout contains a DOT digraph with nodes a, b, and c, three paired internal edges, and two dangling boundary edges. The external edges do not add a cycle, so the triangle's cycle-basis rank remains one.

Verification scope and cost

The docs harness compiles and runs this Rust program; it syntax-checks the setup commands without creating a project or using the network. Success requires `graph.check()` and the subgraph cycle-rank assertion to pass. The run is small; a clean external Cargo build is dependency-dominated and can take minutes, while the graph operation itself should finish in well under a second.

Keep the returned indexes with their graph

a, b, and c are compact `NodeIndex` values owned by this graph construction. Do not serialize their numeric representation as a durable identity or reuse them against another graph. Mutating operations return the mappings needed to relate old and new storage.

2.3 Make the first useful variation

Add a fourth internal edge between a and b, run the program again, and change the assertion to `basis.len() == 2`. This demonstrates the key distinction: parallel paired half-edges add an independent cycle, while dangling identity half-edges describe the graph boundary excluded by `internal`.

From here, use a subgraph view to run traversal, cuts, or contraction against only the region you intend to modify. Enable Linnet's drawing feature only when you need layout rather than plain DOT interchange.

2.4 Troubleshooting and next steps

- A type-inference error around `build()` means Rust cannot choose node storage or data types; keep the explicit `HedgeGraph<&str, &str>` annotation from the example.
- A failed `check()` points to an involution or storage invariant. Validate immediately after parsing or construction, before applying graph algorithms.
- If an algorithm includes an unexpected dangling edge, check that you passed `internal` rather than `graph.full_filter()`, then inspect the `Flow` assigned to each external edge.
- Continue with the subgraph-and-algorithms manual, then consult the Rust API reference for the exact return mappings of mutating operations and additional builder methods.

3 Subgraphs, traversal, cuts, and drawing

Linnet algorithms operate on a graph together with an explicit subgraph view. This lets one algorithm work on the full graph, a selected region, or the boundary created by cutting paired half-edges. Choose the subgraph

representation by operation: bit-backed sets are efficient for repeated set algebra, while borrowed or mapped views avoid copying when identity must remain attached to the owning graph.

3.1 Traversal and connectivity

A traversal needs a starting half-edge or node, the active subgraph, and a policy describing which adjacent half-edges may be crossed. Results retain typed indexes into the original graph. Breadth-first traversal is useful for distance layers and connectivity; depth-first traversal is useful for trees, back edges, and decompositions. A traversal tree is a derived view, not a new graph, so mutations must be coordinated with the indexes it contains.

Connectivity and traversal results depend on the chosen subgraph. Run an operation on the full graph only when external half-edges and excluded nodes should participate in the answer. Linnet does not currently expose a dedicated articulation-point or biconnected-component API; derive those from traversal data only when your application owns that algorithm and its tests.

3.2 Cycles and cuts

`cycle_basis` returns an independent cycle vector together with the spanning forest used to construct it. Use `cyclotomatic_number` when the numerical cycle rank itself is the result. `all_cycles` expands a basis and can grow exponentially; prefer a basis when assigning loop momenta. Symmetric differences combine cycle bitsets without inventing new graph identities.

Cut enumeration separates required left and right node sets and returns the left region, cut content, and right region. A cut edge is represented through its half-edges, preserving which side owns each endpoint. Validate direction/flow after enumeration when a physical Cutkosky or tensor-network interpretation depends on orientation.

Complexity is part of the API choice

Enumerating every cycle or every admissible cut is inherently combinatorial. Filter the subgraph and endpoint sets first, and use basis or connectivity operations when enumeration is not required by the caller.

3.3 Enumerate separating cuts

This complete example builds a square with a diagonal and asks for every cut separating opposite nodes. It is the smallest graph that makes the distinction between a path and a cut family visible without external data or drawing support.

```
use linnet::half_edge::{HedgeGraph, builder::HedgeGraphBuilder};

fn main() {
    let mut builder = HedgeGraphBuilder::new();
    let a = builder.add_node();
    let b = builder.add_node();
    let c = builder.add_node();
    let d = builder.add_node();

    builder.add_edge(a, b, (), true);
    builder.add_edge(b, c, (), true);
    builder.add_edge(c, d, (), true);
    builder.add_edge(d, a, (), true);
    builder.add_edge(b, d, (), true);

    let graph: HedgeGraph<(), (), ()> = builder.build();
    graph.check().expect("the half-edge involution is valid");
    let cuts = graph.all_cuts_from_ids(&[a], &[c]);

    assert_eq!(cuts.len(), 4);
}
```

```
println!("separating cuts: {}", cuts.len());
}
```

The expected invariant is four distinct left/cut/right partitions separating a from c . Changing the diagonal changes that count, while changing only node or edge payloads does not. The generated `HedgeGraph` reference and builder reference give the exact supported type boundaries.

Interpret a cut mismatch structurally

A failed `graph.check()` indicates a storage or involution invariant and must be resolved before enumeration. Zero cuts usually means the endpoint sets overlap or are disconnected from the selected region. An unexpected nonzero count means the constructed edges or endpoint sets do not describe the graph you intended; inspect those before adding post-enumeration filters.

3.4 Mutation boundaries

Extraction copies a selected region, excision separates a graph along its boundary, sewing joins compatible dangling half-edges, and contraction identifies structure. These operations return mappings or replacement indexes where needed. Keep those results: a numeric `Hedge` or `NodeIndex` from before a storage-changing operation is not automatically a valid index into the resulting graph.

3.5 DOT and drawing

DOT parsing and emission are interchange/debugging boundaries. Attributes can carry graph, node, edge, and half-edge data, but the typed Rust invariants are reconstructed by the parser; handle parser errors before running algorithms. The drawing feature enables layout; rendering belongs to a DOT consumer, `Clinnet`, or `Linnest`. Layout coordinates are presentation data and may vary without changing topology, subgraph membership, or edge flow.

Tensor contraction uses Spenso

`Linnet` determines connectivity and graph transformations. Tensor compatibility, contraction cost, and execution belong to `Spenso` even when a `Spenso` network delegates its graph algorithms to `Linnet`.

4 Render DOT figures with Clinnet

`Clinnet` is `Linnet`'s command-line companion for documentation and diagram workflows. The `Cargo` package is named `clinnet`, while the installed executable is named `linnet`. It recursively finds DOT files, renders one PDF per graph through `Typst`, and assembles those PDFs into a grid. Use the Rust library for graph algorithms; use `Clinnet` when the input is already DOT and the desired result is a browsable set of figures.

4.1 Install and render a directory

`Clinnet` requires `Typst` 0.15.0 or newer. Install both commands, then confirm the executable's current flags rather than relying on an older pre-subcommand invocation:

```
cargo install clinnet
cargo install typst-cli --version 0.15.0 --locked
linnet --help
linnet draw examples
```

The final command scans `examples/` recursively. With default paths it creates:

- `build/figs/`, mirroring the input directory and containing one PDF per DOT file;
- `build/grid.pdf`, the combined document;
- `build/templates/`, containing the default figure, grid, layout, `Linnest`, and `Kurvst` assets;

- `build/.cache/figures.json`, the per-figure content cache; and
- `build/.cache/run-metadata.json`, the plan used by targeted redraw commands.

What a second run reuses

A figure is reused only when its DOT source, selected figure template, explicit `--style` files, `--input` values, and the bundled `.typ/.wasm` assets in `build/templates/` have the same content hash. Clinnet does not recursively discover arbitrary imports beside a custom template; pass those files with `--style` when they must invalidate the cache. The grid is still regenerated from the current figure index. Removing a DOT file removes its stale figure output during the next full draw; deleting the figure cache forces every remaining figure to rebuild.

4.2 Customize templates and inputs

Use `--figure-template` for each individual figure and `--grid-template` for the combined PDF. Repeat `--style` for extra files whose contents must invalidate the figure cache. Repeat `--input key=value` to expose strings through Typst's `sys.inputs` in both compilation stages:

```
linnet --build-dir build draw graphs \
  --figure-template templates/figure.typ \
  --grid-template templates/grid.typ \
  --style templates/colors.typ \
  --input theme=dark \
  --input scale=1.5 \
  --output-path build/review.pdf
```

Clinnet supplies `data-path`, `data_path`, and a derived title to the figure template in addition to caller inputs. `--output-path` selects the final grid PDF; it is not a figure-template input. Input values are strings; parse them in Typst when a number or boolean is required. The embedded defaults include the canonical Linnest and Kurvst package trees, so a custom template can use the same graph-layout and curve primitives. Continue to the Linnest Typst API for graph construction, layout, inspection, and drawing rather than copying its wrapper surface into a Clinnet template guide.

Templates must share a Typst project root

Clinnet chooses the narrowest existing directory that contains the template and its inputs. A Typst import outside that tree, a missing style file, or a template moved without its dependencies fails before a PDF is produced. Keep the templates and imported assets beneath one project directory and read the full Typst error printed by Clinnet.

4.3 Make a targeted rebuild

Run a complete draw once so that the build metadata records the input root, templates, output paths, styles, and figures. You can then rebuild one known DOT figure without rescanning the directory, or rebuild only the grid:

```
linnet --build-dir build redraw-figure graphs/box.dot --input theme=light
linnet --build-dir build redraw-grid --output-path build/review.pdf
```

`redraw-figure` starts with the inputs recorded by the full draw and applies new inputs as per-key overrides. `redraw-grid` reuses the cached figure list; if any `--input` is supplied to that command, the supplied list replaces the grid's recorded input list, so repeat every value the grid template still needs. A target not present in the preceding plan must be added with a new full draw.

Use `--no-parallel` for deterministic sequential diagnosis, or `--jobs N` to bound parallel figure compilation. These options affect scheduling, not cache identity or graph semantics.

4.4 Source and release boundaries

The implementation and exact Clap definitions are in the [Clinnet command source](#). Clinnet has its own package version and [version history](#), independent of the Rust `linnet` library. Record the Clinnet and Typst versions when reporting a rendering or cache problem.

5 Linnest Typst API

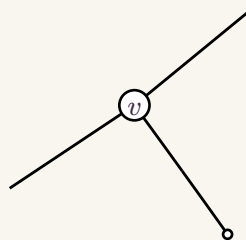
Linnest is the Typst and WebAssembly interface to Linnet’s graph model, DOT parser, layout algorithms, graph queries, and physics-aware drawing styles. This page embeds the package’s complete maintained manual so its workflows and generated module reference are available in the same searchable Linnet documentation corpus.

Choose the interface deliberately

Use Linnet’s Rust or Python interfaces for graph construction and algorithms in an application. Use Linnest when the graph, layout, or final drawing is owned by a Typst document.

5.1 Linnet

The `linnet` crate wrapped by this package is built around a half-edge graph data structure. This means that instead of a graph being represented as a set of nodes and edges, it is represented as a set of half-edges H , and a set of vertices V . The graph structure is then encoded through two maps. The first map, $\partial : H \rightarrow V$ maps each half-edge to its corresponding vertex. The preimage of any vertex v is the set of half-edges that map to it, called the crown of v . The second map, $\iota : H \rightarrow H$, is an involution that *glues* half-edges together to form edges. If a half-edge is glued to itself, we call that an external half-edge. This means that linnet graphs are strictly more capable than normal edge and vertex graphs.



Native half-edges also make subgraphs more granular because they can be encoded as sets of half-edges. This directly supports vertex-induced subgraphs: the union of the crowns of a set of vertices.

5.2 Linnest

Linnest is the Typst and WebAssembly interface to Linnet. It provides layout algorithms, DOT parsing, and selected graph algorithms without a separate runtime process.

Graphs can be constructed in two ways: parse a DOT string with `parse`:

```
#let g = parse("digraph { a -> b }")
```

or build from edges and nodes with `build`, using a Fletcher-inspired syntax:

```
#let g = build({
  node(<a>, label: [$v$])
  node(<b>)
  edge(source(<a>), <a-b>, sink(<b>), label: [e])
  edge(source(<a>), <in-a1>, label: [e])
  edge(source(<a>), <in-a2>, label: [e])
})
```

In either case, `type(g)` is dictionary: graph values wrap an archived Linnet graph together with native Typst data. Rust owns topology, layout state, statement metadata, and internal opaque payload bytes. User data captured from Typst stays in Typst and is merged back into query records.

The main use case is to place nodes and edges on a canvas with the `layout` function and render the result with `draw`.

- graph for construction, parsing, inspection, joins, and graph algorithms.
- subgraph for subgraph object construction and inspection.
- layout for the separate layout pass.
- draw for rendering a laid-out graph object with CeTZ.
- physics for reusable particle-line edge styles.

5.2.1 Choose an import path

Linnest and Kurvst are currently bundled source packages, not Typst Universe packages. A Clinnet run writes both package trees below `build/templates/`. From a custom template in that directory, import Linnest with:

```
#import "crates/linnest/typst/src/lib.typ": draw, graph, layout, subgraph
```

From this repository's `crates/linnest/typst/examples/` directory, the equivalent checkout-relative import is `../src/lib.typ`. Keep the package directory and its `linnest.wasm` file together when copying it elsewhere. The examples below use the checkout-relative form because they are also compiled as repository tests.

5.2.2 Minimal Build Example

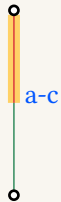
```
#import "../src/lib.typ": draw, graph, layout, subgraph
#import graph: build, dot, edge, edges, node, nodes, parse, sink, source

#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>, compass: "e"),
    <a-c>,
    sink(<c>, compass: "w"),
    label: [a-c],
    statements: (
      color: "0055ff",
      source-color: "d72638",
      sink-color: "1b7f4c",
    ),
  ),
},
  name: "demo",
)
#let g = layout(g)
#let east = subgraph.compass(g, "e")
#let edge-records = edges(g, subgraph: east)
#let dot-text = dot(g)
#let edge-label(edge) = text(fill: rgb("#" + edge.color))[#edge.label]
#let source-style(edge) = (stroke: rgb("#" + edge.source-color) + 0.5pt)
#let sink-style(edge) = (stroke: rgb("#" + edge.sink-color) + 0.5pt)
#context if target() == "paged" {
  draw(
    g,
    subgraph: east,
    edge-label: edge-label,
    edge-label-style: (anchor: "south"),
    source-style: source-style,
```

```

    sink-style: sink-style,
  )
} else {
  [The downloadable PDF renders this CeTZ result. The HTML manual keeps the
  copyable source because Typst's experimental HTML target does not yet emit
  the drawing content.]
}

```



5.2.3 Graph Objects

Graph objects are Typst dictionaries wrapping archived Rust graph bytes plus native Typst data arrays for graph, node, edge, source, and sink data. The Rust graph may also carry an internal opaque payload, but that payload is used only by the Typst wrapper and is not exposed in public records. Build or parse graph objects with `graph`, transform graph objects with `layout`, and pass objects back to `graph` or `subgraph` for inspection. Subgraph objects are still opaque zero-copy values.

- `graph.parse(input)` parses one or more DOT digraphs and returns an array of graph objects. Its `eval-graph-fields`, `eval-node-fields`, `eval-edge-fields`, `eval-source-fields`, and `eval-sink-fields` arguments are convenience arguments for `graph.eval-fields`.
- `graph.build(..)` constructs one graph object from a stream of node and edge items.
- `graph.map(graph, ..)` maps graph, node, edge, source, and sink records to new native data without changing topology.
- `graph.node-data(graph, <name>)` and `graph.edge-data(graph, <name>)` return one named node or edge data.
- `graph.update-node-data(graph, <name>, data)` and `graph.update-edge-data(graph, <name>, data)` update one named node or edge data. Direct replacements and callbacks run in Typst with `(data, record)`.
- `graph.eval-fields(graph, ..)` evaluates selected record fields into data entries. It works on parsed and built graph objects.
- `node(..)` returns a node item.
- `source(..)` and `sink(..)` return half-edge endpoints.
- `edge(..)` returns an edge item built from source/sink endpoints and an optional edge name.
- `layout(graph, ..)` runs layout as an explicit second step. Its settings are named parameters so calls stay descriptive and Tidy can document each field.
- `draw(graph, ..)` draws a laid-out graph object with CeTZ.
- `dot(graph)` returns a DOT string for inspection or export.

5.2.4 Graph Specs

The Typst construction API is half-edge first: create node items, create source and sink half-edge endpoints, then pass edge items to `graph.build`. `graph.build` accepts both comma-separated items and ordinary Typst code blocks. Typst labels such as `<a>`, `<h1>`, and `<el>` are API names; node and edge names are emitted back from `nodes(g)` and `edges(g)` as Typst labels. They are resolved before the wire format is sent to the Rust plugin. Numeric id arguments choose graph indexes or ordering: on nodes, id fixes the resulting node index and must be unique and in bounds. On nodes, edges, sources, and sinks, extra named arguments are captured as opaque Typst data fields: `edge(source(<a>, style: physics.source-stroke()), sink(<b>), particle: "g")` stores `(style: ..)` in the source data and `(particle: "g")` in the edge data. These user data values are not sent through the Rust plugin boundary. Typst sends an internal opaque payload with correlation data, asks Rust to resolve the graph indexes, then stores user data in arrays at those resolved graph, node, edge, and half-edge ids. `graph.info`,

`graph.nodes`, and `graph.edges` merge those native values into the returned records. A captured label data field on nodes and edges is display content used by the default drawing style; use statements: (`label: "..."`) when a flat metadata label string is needed. Statements are flat metadata used by DOT; they cannot nest. Values are scalar strings/numbers/booleans. Use data fields for structured Typst data or content.

`graph.build` and `graph.parse` also accept `default-node-data`, `default-edge-data`, `default-source-data`, and `default-sink-data`. These defaults are merged into the corresponding data; captured data fields on nodes, edges, sources, and sinks override the defaults. For drawing, the physics helpers read `data.style` on source and sink half-edges and `data.display-label/data.label` on edges:

```
#let g = build({
  node(<a>, label: [a])
  node(<b>, label: [b])
  edge(
    source(<a>, style: physics.source-stroke(c: red)),
    <e>,
    sink(<b>, style: physics.sink-stroke(c: blue)),
    label: [$p$],
    particle: "g",
  )
},
  default-edge-data: (kind: "propagator"),
)
#let callbacks = physics.style()
#draw(
  layout(g),
  source-style: callbacks.source-style,
  sink-style: callbacks.sink-style,
  edge-label: callbacks.edge-label,
)
```

When parsing DOT, the same native data arrays can be filled from selected string fields. Rust parses DOT into topology and statement metadata; Typst applies defaults and evaluates selected fields afterward. The selected fields are evaluated with the record's merged fields dictionary in scope. Since node names are labels, use `#str(name)` when a name should become visible text:

```
#let g = parse(
  "digraph g { a [label=\\"A\\"]; a -> b [label=\\"$p$\\", source=\\"out\\", sink=\\"in\\"] }",
  eval-node-fields: ("label"),
  eval-edge-fields: ("label"),
  eval-source-fields: ("statement"),
  eval-sink-fields: ("statement"),
).first()
#nodes(g).first().data.label
#edges(g).first().data.label
```

The same transform can run after construction. This is useful for global edge statements that should apply to every edge while still seeing local edge fields:

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), sink(<b>), statements: (mom: "p"))
}, default-edge-statements: (display-label: "$#mom$"))
#let g = graph.eval-fields(g, eval-edge-fields: ("display-label"))
#edges(g).first().data.at("display-label")
```

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(source(<a>), <e1>, sink(<c>))
})
```

Named nodes and edges can be updated after construction without scanning in the caller. The update replaces the native data, or it can be a callback receiving (data, record):

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(source(<a>), <e1>, sink(<c>))
})
#let g = update-node-data(g, <a>, (label: [A]))
#let g = update-edge-data(g, <e1>, (data, edge) => (
  label: [$p$],
  source: edge.source.node,
))
#node-data(g, <a>).label
#edge-data(g, <e1>).label
```

source(..) and sink(..) accept a node reference plus optional name, id, statement, and compass. name is a Typst label name for the half-edge; id is numeric:

```
edge(source(<a>, name: <h1>, id: 0), <e1>, sink(<c>, id: 2), label: [a-c])
```

One half-edge creates an external edge. The side is determined by the constructor, so there is no public flow argument:

```
edge(<incoming>, sink(<a>))
edge(source(<c>), <outgoing>)
```

graph.build does not interpolate statement strings on the Rust side. Use graph.eval-fields or graph.map when a default statement should turn into Typst content or structured data. The evaluation scope includes the record's merged fields, so default edge statements can still refer to local edge fields:

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <a-c>,
    sink(<c>),
    label: [a-c],
    statements: (color: "0055ff", label: "a-c"),
  )
},
default-edge-statements: (
  color: "000000",
  display-label: "$#label$",
),
)
#let g = graph.eval-fields(g, eval-edge-fields: ("display-label",))
```

5.2.5 Placements

`graph.pos` creates a first-class placement. The default mode: "pin" turns a coordinate into a fixed layout constraint and a drawable position. Use mode: "start" when the coordinate should only seed the layout:

```
#let g = build({
  node(<a>, pos: graph.pos(x: -2, y: 0))
  node(<c>, pos: graph.pos(ref: <a>, dx: 4, dy: 0))
  edge(source(<a>), <a-c>, sink(<c>), pos: graph.pos(x: 0, y: 1.2))
})
```

`graph.group` links one coordinate across several nodes or edge control points. A side of "+" keeps the coordinate positive, and "-" keeps it negative. GammaLoop external-edge columns use this to keep incoming and outgoing external legs on opposite sides while pairing rows by a shared y group:

```
#let edge-items = (
  edge(
    source(<right-ext>),
    sink(<center>),
    pos: graph.pos(x: graph.group("right", side: "+"), y: graph.group("edgee0")),
  ),
  edge(
    source(<center>),
    sink(<left-ext>),
    pos: graph.pos(x: graph.group("left", side: "-"), y: graph.group("edgee0")),
  ),
)
```

Raw DOT input uses the same placement model through the `pos` attribute. The standard Graphviz subset is preserved: `pos="x,y"` is a starting coordinate and `pos="x,y!"` is pinned. Linnest extends the value with explicit id references and axis entries. In axis entries, ! belongs to that axis: numeric entries without ! are starting coordinates, numeric entries with ! are fixed constraints, and grouped entries must use ! because groups are constraints.

```
digraph {
  a [id=0 pos="0,0!"]
  b [id=1 pos="ref(node:0)+4,0!"]
  a -> b [id=0 pos="ref(node:1)+0,1!"]
  b -> c [id=1 pos="ref(edge:0)+1,0!"]
  c [id=2 pos="x:2!"]
  d [id=3 pos="x:2!,y:1"]
  ext -> a [id=2 pos="x:@-left!,y:@edge0!"]
}
```

The DOT grammar is intentionally id-based: `ref(node:0)` uses a node id and `ref(edge:0)` uses an edge id. Bare names and implicit edge order are not placement references. The @ syntax denotes grouped coordinate constraints; @+name and @-name keep the grouped coordinate on the positive or negative side respectively.

```
pos="x,y"           // start x and y
pos="x,y!"          // pin x and y
pos="x:<coord>"      // start numeric x
pos="x:<coord>!"     // pin x
pos="y:<coord>!"     // pin y
pos="x:<coord>!,y:<coord>" // pin x, start numeric y
pos="x:<coord>,y:<coord>!" // start numeric x, pin y
```

Draw styling is Typst-native. Pass dictionaries or callbacks to `draw`; edge callbacks receive the merged scope, edge statements, source/sink half-edge records, and the edge index:

```
#let edge-label(edge) = text(fill: rgb("#" + edge.color))[#edge.display-label]
#let source-style(edge) = (stroke: red + 0.5pt)
#let sink-style(edge) = (stroke: blue + 0.5pt)
#draw(layout(g), edge-label: edge-label, source-style: source-style, sink-style: sink-style)
```

Edge labels are drawn with CeTZ content at the layout label position. Use `edge-label-style: (anchor: "south")`, or an edge-data callback returning a style dictionary, to choose which point of the label is anchored there.

Marks can follow graph orientation as a first-class draw option. Put the same mark layer on both halves and set `mark-orientation: "edge"`; default-oriented edges mark the source half, reversed edges mark the sink half with the marker flipped, and undirected edges suppress the mark.

```
#let oriented-arrow = (
  stroke: black + 0.7pt,
  mark: (end: (symbol: ">", fill: black, anchor: "center", shorten-to: auto), scale: 0.75),
  mark-position: "center-if-dangling",
  mark-orientation: "edge",
)
#draw(layout(g), source-style: oriented-arrow, sink-style: oriented-arrow)
```

5.2.6 Physics Edge Styles

`physics.style(..)` returns `source-style`, `sink-style`, and `edge-label` callbacks for `draw`. The default source/sink strokes use darker/lighter halves to encode the graph source/sink split. Fermion map entries marked with `fermion-flow` receive one particle-flow arrow on the main edge using `mark-orientation: "edge"`; paired edges follow `edge.orientation`, dangling half edges place the arrow in the middle of the visible half edge, and undirected edges omit the arrow. This is independent of momentum arrows.

Set `momentum-arrows: true` to draw centered parallel arrow decorations while the main edge remains connected to the nodes. The arrows flow from source to sink independently of the DOT `dir/orientation` value; there is one momentum marker per edge, even though the base edge is internally styled as source and sink halves. On paired edges the marker lives on the sink half; on dangling edges it lives on the existing source or sink half-edge. The decoration defaults to a plain black 0.55pt stroke and a scaled CeTZ "barbed" arrowhead, so it does not inherit the base particle stroke. The arrow length is capped by both `momentum-arrow-length` and `momentum-arrow-ratio`, so the shorter one wins. When the layout provides an edge-label position, the arrow offset is signed so the momentum marker is drawn on the same side of the edge as that label. Use `momentum-arrow-offset` to set the normal displacement. `momentum-arrow-mark: auto` uses the default barbed marker, `momentum-arrow-mark: none` draws only the momentum line, and any other value overrides the CeTZ mark style, for example "stealth" or `(end: "barbed", scale: 1.6)`. Only the end mark is kept so source/sink splitting cannot create a second momentum head.

Optional labels can be built from edge metadata with `show-edge-index`, `show-half-edge-index`, `show-particle`, and, for explicit momentum fields, `show-momentum`. `show-half-edge-index` only emits a value for dangling half edges.

```
#import "../src/lib.typ": draw, graph, layout, physics
#import graph: build, dot, edge, edges, node, nodes, parse, sink, source

#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <fermion-main>,

```

```

    sink(<c>),
    id: 7,
    particle: "fermion",
  )
  edge(
    source(<c>),
    <fermion-out>,
    id: 8,
    particle: "fermion",
  )
},
name: "physics",
)
#let callbacks = physics.style(
  momentum-arrows: true,
  show-edge-index: true,
  show-particle: true,
)
#draw(
  layout(g),
  source-style: callbacks.source-style,
  sink-style: callbacks.sink-style,
  edge-label: callbacks.edge-label,
)

```

Set edge-offset on draw, or offset in a source-style/sink-style dictionary, to draw a fitted parallel path. Style callbacks may also return an array of dictionaries; the layers are drawn in order on the same graph edge, so parallel strokes do not require duplicate graph edges.

```

#let base-style(edge) = (
  stroke: (paint: gray, thickness: 0.7pt, cap: "round"),
)
#let offset-style(edge) = (
  offset: 0.18,
  length: 1.4,
  ratio: 0.5,
  resolve-length: "min",
  stroke: (paint: rgb("#2f6f4e"), thickness: 1pt, cap: "round"),
)
#let source-style(edge) = (base-style(edge), offset-style(edge))
#let sink-style(edge) = (base-style(edge), offset-style(edge) + (mark: (end: ">")))

```

The parallel path is applied to the base edge geometry before patterns and other decorations; node outsets then trim the shifted path, so shifted paths still start and end outside fitted node circles. Add edge-length or length to center-trim the shifted path to a fixed arc length, and add edge-ratio or ratio to cap it by a fraction of the base edge length. edge-resolve-length

resolve-length decides how to combine both limits "min"/"shorter" (default), "max"/"longer",

"length"/"fixed", "ratio"/"relative", "none"/"full", or a function receiving (base-length, length, ratio). Set offset-side: "label" on an offset layer to choose the sign of offset so the layer is drawn on the same side of the curve as the edge label.

Data defaults are also the global styling hook for all sources, sinks, nodes, and edges. More specific data can be added with captured named arguments on node, edge, source, and sink items, or by running graph.map/graph.eval-fields after construction. This keeps evaluated Typst values in the native data channel instead of adding renderer-specific eval fields to the Rust topology spec:

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <a-c>,
    sink(<c>),
    label: [a-c],
    kind: "highlight",
  )
},
  name: "demo",
  default-source-data: (style: (stroke: red + 0.5pt)),
  default-sink-data: (style: (stroke: blue + 0.5pt)),
)
#let callbacks = physics.style()
#draw(layout(g), source-style: callbacks.source-style, sink-style: callbacks.sink-style)
```

graph.build also accepts comma-separated items:

```
#let g = build(
  node(<a>),
  node(<b>),
  edge(
    source(<a>, compass: "e"),
    <ab>,
    sink(<b>, compass: "w"),
    label: [ab],
    statements: (color: "0055ff", label: "ab"),
  ),
  name: "demo",
  statements: (full_num: "x + y"),
  default-node-statements: (shape: "circle"),
  default-edge-statements: (
    color: "000000",
    display-label: "$#label$",
  ),
)
```

5.2.7 Graph Queries

graph.info(g) returns graph metadata. nodes(g) returns node records, and edges(g) returns edge records. Node and edge record name values are Typst labels when present. Pass subgraph: sg to filter nodes or edges by a subgraph object.

graph.join(left, right, key: "statement") joins matching dangling half edges. The key is read from half-edge statements or numeric ids and can be "statement", "compass", or "id".

graph.cycles(g) returns subgraph objects for a cycle basis. graph.forests(g) returns subgraph objects for spanning forests.

5.2.8 Layout Model

layout starts from a traversal-tree placement, then optimizes the positions of graph nodes and edge control points. The initial tree spacing is

$$L = \lambda \sqrt{\frac{WH}{\max(n, 1)}}$$

with horizontal spacing $\tau_x L$ and vertical spacing $\tau_y L$. Here λ is length-scale, W is viewport-w, H is viewport-h, τ_x is tree-dx, and τ_y is tree-dy. These fields set the geometry scale for both layout modes.

For deterministic, non-iterative placement, use `layout-algo: "tree"` or `layout-algo: "dot"`. "tree" places a traversal forest by levels. "dot" uses a directed layered placement for acyclic inputs, falling back to tree placement when the selected edges contain a cycle. Both modes also work on a subgraph:

```
#let g = parse("digraph partial { a -> b; b -> c; c -> d; d -> a }").at(0)
#let tree = graph.forests(g).at(0)
#let g = layout(g, layout-algo: "tree", subgraph: tree)
#draw(g)
```

Only nodes touched by the selected subgraph are placed. Edge control points are then resolved from the current node positions, so edges outside the selected subgraph are drawn as straight lines unless their own position constraints say otherwise.

`layout-algo: "force"` and `layout-algo: "anneal"` also accept `subgraph`. For these iterative modes, nodes and edge control points outside the selected subgraph stay fixed and act as boundary points while the selected subgraph is optimized.

Set `layout-nodes: "fixed"` to keep every node at its current `pos` for this layout pass and move only edge control points. With `subgraph`, only edges in the selected subgraph are moved; all other edge control points keep their current positions. The fixed-node policy is temporary; the returned graph stores the resulting coordinates as `pos`, but it does not turn them into persistent `pin` constraints. This is useful after one node-placement pass when a later pass should route or relax selected edges without disturbing the node layout:

```
#let g = parse("digraph partial { a [pos=\"0,0\"]; b [pos=\"4,0\"]; c [pos=\"8,0\"]; a -> b; b -> c }").at(0)
#let first = subgraph.bits(g, (true, true, false, false))
#let g = layout(g, layout-algo: "tree", layout-nodes: "fixed", subgraph: first)
#draw(g)
```

The shared spring/charge model uses the following coefficients:

$$c_{vv} = \beta L^2$$

$$c_{ev} = \beta \gamma_{ev} L^2$$

$$c_{ee} = \beta \gamma_{ee} L^2$$

$$c_{\text{center}} = \beta g_{\text{center}} L^2$$

$$c_{\text{dangling}} = \beta \gamma_{\text{dangling}} L^2$$

The Typst parameter names are β for `beta`, γ_{ev} for `gamma-ev`, γ_{ee} for `gamma-ee`, g_{center} for `g-center`, and γ_{dangling} for `gamma-dangling`. The spring stiffness k is `k-spring`, and the softening constant ε is `eps`.

In `layout-algo: "anneal"`, `linnest` minimizes an energy:

$$E = \sum_{i < j} \frac{1}{2} \frac{c_{vv}}{d(v_i, v_j) + \varepsilon} + \sum_{i, e} \frac{c_{ev}}{d(v_i, e) + \varepsilon} + \sum_{(v, e) \text{ incident}} \frac{1}{2} k (\ell_e - d(v, e))^2 \\ + \sum_{\text{local edge pairs}} \frac{1}{2} \frac{c_{ee}}{d(e_i, e_j) + \varepsilon} + \sum_{\text{dangling pairs}} \frac{1}{2} \frac{c_{\text{dangling}}}{d(e_i, e_j) + \varepsilon} + \sum_i \frac{1}{2} c_{\text{center}} d(v_i, 0)^2 + p_{\text{cross}} N_{\text{cross}}$$

Here p_{cross} is crossing-penalty and N_{cross} is the number of detected edge crossings. The quadratic center term pulls nodes toward the origin at every radius. `temp`, `step`, `seed`, `steps`, `epochs`, `cool`, `accept-floor`, `step-shrink`,

and incremental-energy belong to this simulated annealing mode. crossing-penalty is also anneal-only; force mode does not currently add a crossing force.

In layout-algo: "force", linnest applies the direct forces corresponding to the same vertex-vertex, edge-vertex, incidence spring, local edge-edge, dangling-edge, and center terms. step is the integration step, delta clamps per-step movement, steps and epochs set the iteration budget, cool shrinks the step after each epoch, and early-tol stops when movement is small. z-spring and z-spring-growth are force-only helpers: the integrator gives points temporary z coordinates to break overlaps and pulls them back toward the 2D plane.

directional-force is applied in both modes as an extra bias derived from pin/port direction constraints.

After either graph layout mode, labels are relaxed separately. If L_l is the label target distance and q_l is the label repulsion strength, then $L_l = \alpha_l L$ and $q_l = \beta_l L^2$. The Typst names are label-length-scale for α_l and label-charge for β_l . label-spring is the spring constant pulling each label toward its target. label-layout: "normal" uses a perpendicular offset target. With label-layout: "dangling-tangent", paired edges still use that perpendicular target, but dangling half-edge labels are offset along the edge direction away from the attached node. With label-layout: "fixed-length", the label remains at distance L_l from the edge point and only rotates around it under repulsive forces. label-steps, label-step, label-early-tol, and label-max-delta-scale control the label relaxation iteration.

5.2.9 Subgraphs

Subgraph objects are opaque zero-copy values.

- `subgraph.label(g, label)` constructs a subgraph from a base62 label.
- `subgraph.bits(g, bits)` constructs a subgraph from a boolean hedge array.
- `subgraph.compass(g, compass)` selects half edges with a DOT compass point.
- `subgraph.to-label(sg)` returns the base62 label.
- `subgraph.hedges(sg)` returns included hedge indices.
- `subgraph.contains(sg, hedge)` tests hedge membership.

5.2.10 Generated Reference

5.3 graph

- `build()`
- `parse()`
- `node()`
- `source()`
- `sink()`
- `edge()`
- `group()`
- `pin()`
- `start()`
- `pos()`
- `map()`
- `style()`
- `eval-fields()`
- `info()`
- `dot()`
- `nodes()`
- `edges()`
- `node-data()`
- `edge-data()`
- `update-node-data()`
- `update-edge-data()`
- `join()`

- `cycles()`
- `forests()`

5.3.1 build

Build one graph object from a stream of node and edge items.

Use `node`, `source`, `sink`, and `edge` to create graph items. Positional items may be passed as comma-separated arguments or yielded from a Typst code block.

5.3.1.1 Parameters

```
build(
  ..items: array,
  name: none string,
  data: any,
  statements: dictionary,
  default-node-statements: dictionary,
  default-edge-statements: dictionary,
  default-node-data: any,
  default-edge-data: any,
  default-source-data: any,
  default-sink-data: any
) -> dictionary
```

5.3.1.1.1 ..items array

Node and edge items returned by `node` and `edge`. The best way to use these is to use a code scope, so that the edges and nodes append each other:

```
#let g = build({
  node(<a>, label: [a])
  node(<b>, label: [b])
  node(<c>, label: [c])
  edge(source(<a>), sink(<b>))
  edge(source(<b>), sink(<c>))
  edge(source(<a>), sink(<c>))
})
>>>#align(center+horizon, draw(layout(g)))
```

5.3.1.1.2 name none or string

Graph name.

```
#let g = build(name: "My Graph", {
})
#info(g).name
```

Default: `none`

5.3.1.1.3 data any

Native Typst graph data.

```
#let g = build(data: (a:(b:(1, ))), {  
  })  
#info(g).data
```

Default: `none`

5.3.1.1.4 statements dictionary

Flat graph statements, that get turned into a string to string dictionary in rust. Used by DOT parsing; values cannot nest.

```
#let g = build(statements: (a:1), {  
  })  
#info(g).global-statements
```

Default: `(:)`

5.3.1.1.5 default-node-statements dictionary

Flat default node statements. Used by DOT; values cannot nest.

Default: `(:)`

5.3.1.1.6 default-edge-statements dictionary

Flat default edge statements. Used by DOT parsing; values cannot nest. These are applied/delegated to all edges.

```
#let g = build(default-edge-statements: (a:1), {  
  node(<a>)  
  edge(source(<a>))  
  edge(sink(<a>))  
})  
#edges(g).map(e=>e.statements)
```

Default: `(:)`

5.3.1.1.7 default-node-data any

Default data merged into every node data. Captured node data fields override it.

```
#let g = build(default-node-data: (a:1), {  
  node(<a>)  
  node(<b>,a:2)  
  node(<c>,b:(a:1))  
})  
#nodes(g).map(n=>n.data)
```

Default: **none**

5.3.1.1.8 default-edge-data any

Default data merged into every edge data. Captured edge data fields override it.

```
#let g = build(default-edge-data: (a:1,b:[ $m_\mu$ ])), {  
  node(<a>,id:0)  
  edge(source(<a>),id:1)  
  edge(sink(<a>),<e>,id:0,b:[set text(font:"Reforma")  
    This is a test: ])  
})  
#edges(g).map(e=>e.data.b).join()
```

Default: **none**

5.3.1.1.9 default-source-data any

Default data merged into every source half-edge data. Captured source data fields override it.

```
#let g = build(default-source-data: (a:1), {  
  node(<a>)  
  edge(source(<a>))  
  edge(sink(<a>))  
})  
#edges(g).map(e=>if e.source != none {e.source.data} else {none})
```

Default: **none**

5.3.1.1.10 default-sink-data any

Default data merged into every sink half-edge data. Captured sink data fields override it.

```
#let g = build(default-sink-data: (a:1), {
  node(<a>)
  edge(source(<a>))
  edge(sink(<a>))
})
#edges(g).map(e=>if e.sink != none {e.sink.data} else {none})
```

Default: `none`

5.3.2 parse

Parse one or more DOT graphs into graph objects.

Default data are applied before `eval-*` fields are evaluated, so default data strings can refer to parsed record fields such as `#str(name)`. Parsed fields take precedence over default data fields, so DOT `label="..."` overrides `default-node-data: (label: ...)`.

Linnest uses dot-parser for DOT syntax and then gives special meaning to a small set of attributes. Every other attribute is preserved as a flat string statement. Preserved statements are visible through `info`, `nodes`, `edges`, `map`, and `eval-fields`, but they do not become native Typst data unless a matching `eval-*` argument is passed.

```
#let src = ``dot
digraph { a [label="A"]; }
``

#let n = nodes(parse(src.text).first()).first()
#n.statements.label
#n.data
```

Graph-level handling:

- The DOT graph name becomes `graph.info(g).name`. A graph-level name attribute can name an anonymous graph; a named graph/digraph header takes precedence.

```
#let src = ``dot
digraph header_wins { graph [name="ignored"]; a }
``

#info(parse(src.text).first()).name
```

- Top-level `key=value` statements and `graph [key=value]` attributes become `graph.info(g).global-statements`, except for the consumed name. They do not configure layout, even when a key looks like a layout option; pass layout options directly to `layout`.

```
#let src = ``dot
digraph { steps=1; graph [subtitle="nested"]; a }
``

#let statements = info(parse(src.text).first()).global-statements
```

```
#statements.steps
#statements.subtitle
```

- node [key=value] and edge [key=value] become default-node-statements and default-edge-statements. They are merged into each parsed node or edge before local attributes, so local attributes override defaults.

```
#let src = ```dot
digraph { node [color=gray]; edge [particle=g]; a [color=red]; a -> b }
```

#let g = parse(src.text).first()
#nodes(g).first().statements.color
#edges(g).first().statements.particle
```

Node handling:

- The DOT node id becomes the node name returned by nodes. If the DOT node id is numeric, it is consumed as the node index instead and the public name is none.

```
#let src = ```dot
digraph { alpha; 1; }
```

#nodes(parse(src.text).first()).map(n => n.name)
```

- id=<n> is also consumed as an explicit node index. Explicit node indexes must be unique and in bounds after parsing.

```
#let src = ```dot
digraph { a [id=1]; b [id=0]; }
```

#nodes(parse(src.text).first()).map(n => n.name)
```

- style=invis marks the DOT node as a dangling external endpoint. It is not returned by nodes. Its remaining attributes are copied onto the external edge that touches it, except shape and label; style and id are consumed.

```
#let src = ```dot
digraph { ext [style=invis, column=left, label=skip]; ext -> a [id=0]; }
```

#let g = parse(src.text).first()
#nodes(g).map(n => n.name)
#edges(g).first().statements.column
#edges(g).first().statements.at("label", default: none)
```

- Node style is consumed for every node; only style=invis has special behavior. Use another attribute name for drawing style metadata that must survive parsing.

```
#let src = ```dot
digraph { a [style=filled, "draw-style=filled"]; }
```

#nodes(parse(src.text).first()).first().statements
```

- pos is parsed as node placement. Simple numeric values are exposed as node.pos; extended placement values are used by layout. pin is an explicit layout constraint and is not exposed as a public statement.

```
#let src = ``dot
digraph { a [id=0, pos="0,0!"]; b [id=1, pos="ref(node:0)+2,0!"]; }
```


```
#let parsed-nodes = nodes(parse(src.text).first())
#parsed-nodes.map(n => n.pos)
#parsed-nodes.map(n => n.statements.at("pin", default: none))
```


```

- shift is parsed as a drawing shift and also remains available as a statement. eval is retained for legacy round-tripping. Other node attributes are preserved as node statements.

```
#let src = ``dot
digraph { a [shift="0.1,0", eval=legacy, label=A]; }
```


```
#let n = nodes(parse(src.text).first()).first()
#n.shift
#n.statements
```


```

Edge handling:

- id=<n> is consumed as the edge index. It is not preserved as a statement; drawing callbacks expose the resulting stable edge index as eid. Explicit edge indexes must be unique and in bounds. Without explicit ids, edge order is parser-internal, not DOT input order.

```
#let src = ``dot
digraph { a -> b [id=1, label=later]; b -> c [id=0, label=first]; }
```


```
#let parsed-edges = edges(parse(src.text).first())
#parsed-edges.map(e => e.edge)
#parsed-edges.map(e => e.statements.label)
#parsed-edges.map(e => e.statements.at("id", default: none))
```


```

- dir=forward, dir=back, and dir=none become edge orientations "default", "reversed", and "undirected". If omitted, directed DOT edges are "default" and undirected DOT edges are "undirected".

```
#let src = ``dot
digraph { a -> b [id=0]; b -> c [id=1, dir=back]; c -> d [id=2, dir=none]; }
```


```
#edges(parse(src.text).first()).map(e => e.orientation)
```


```

- source="..." and sink="..." are consumed as endpoint statement values and removed from edge statements. On an external edge, use the attribute matching the real endpoint side in the DOT edge: ext -> a uses sink=..., while a -> ext uses source=....

```
#let src = ``dot
digraph { ext [style=invis]; ext -> a [id=0, sink=in]; a -> ext [id=1, source=out]; }
```


```
#let endpoint-statement(endpoint) = if endpoint == none { none } else
{ endpoint.at("statement", default: none) }
#let parsed-edges = edges(parse(src.text).first())
#parsed-edges.map(e => endpoint-statement(e.source))
#parsed-edges.map(e => endpoint-statement(e.sink))
#parsed-edges.map(e => e.statements.at("source", default: none))
```


```

- pos is parsed as the edge control-point placement. pin is an explicit edge layout constraint. shift, label-pos, label-angle, and bend are parsed into edge geometry fields; except for pin, they also remain available as statements. Other edge attributes are preserved as edge statements.

```
#let src = ``dot
digraph { a -> b [id=0, pos="0,1!", pin="x:@edge", shift="0.1,0", "label-
pos"="0,1.2", "label-angle"="0.3rad", bend="0.4rad", particle=g]; }
``

#let e = edges(parse(src.text).first()).first()
#e.pos
#e.shift
#e.label-pos
#e.label-angle
#e.bend
#e.statements.particle
#e.statements.at("pin", default: none)
```

- Attributes copied from an invisible endpoint node are merged into the external edge statements unless their keys are shape or label.

```
#let src = ``dot
digraph { ext [style=invis, column=left, shape=none, label=skip]; ext -> a [id=0]; }
``

#edges(parse(src.text).first()).first().statements
```

Port and half-edge handling:

- DOT ports of the form node:port and node:port:compass are preserved on source/sink endpoint records. Numeric ports also assign explicit half-edge ids. Non-numeric ports are exposed only as port-label.

```
#let src = ``dot
digraph { a:left:e -> b:0:w [id=0]; }
``

#let e = edges(parse(src.text).first()).first()
#e.source.port-label
#e.sink.hedge
```

- Compass values are exposed as compass; valid DOT compass names include n, ne, e, se, s, sw, w, nw, c, and _.

```
#let src = ``dot
digraph { a:n -> b:sw [id=0]; }
``

#let e = edges(parse(src.text).first()).first()
#e.source.compass
#e.sink.compass
```

- Explicit half-edge ids from numeric ports must be unique and in bounds. They are retained for half-edge ordering and for join with key: "id"; query and eval records expose the resulting half-edge index as hedge.

```
#let src = ``dot
digraph { a:1 -> b:0 [id=0]; }
``

#let e = edges(parse(src.text).first()).first()
#e.source.hedge
#e.sink.hedge
```

Placement fields:

- `pos="x,y"` gives a starting coordinate; `pos="x,y!"` pins both axes.

```
#let src = ```dot
digraph { a [id=0, pos="0,0"]; b [id=1, pos="2,0!"]; }
```

#let parsed-nodes = nodes(parse(src.text).first())
#parsed-nodes.map(n => n.pos)
#parsed-nodes.map(n => n.statements.at("pos-mode", default: none))
```

- `pos="ref(node:<id>)+dx,dy!"` and `pos="ref(edge:<id>)+dx,dy!"` reference explicit DOT node or edge ids, not names and not implicit parse order.

```
#let src = ```dot
digraph { a [id=0, pos="1,1!"]; b [id=1, pos="ref(node:0)+2,0!"]; a -> b [id=0,
pos="ref(node:1)+0,1!"]; }
```

#let g = parse(src.text).first()
#nodes(g).at(1).pos
#edges(g).first().pos
```

- Axis form accepts `x:<coord>` and `y:<coord>` entries. `!` applies to the individual axis, for example `pos="x:2!,y:0"`.

```
#let src = ```dot
digraph { a [id=0, pos="x:2!,y:0"]; }
```

#nodes(parse(src.text).first()).first().pos
```

- Group coordinates use `@name`, `@+name`, or `@-name` in axis form and must be pinned with `!`, for example `pos="x:@-left!,y:@row!"`.

```
#let src = ```dot
digraph { ext [style=invis]; ext -> a [id=0, pos="x:@-left!,y:@row!"]; }
```

#edges(layout(parse(src.text).first(), layout-algo: "tree")).first().pos
```

- `pin` accepts numeric point constraints, `x:<coord>`, `y:<coord>`, and grouped constraints such as `x:@left`, `x:@+right`, or `@row`.

```
#let src = ```dot
digraph { a -> b [id=0, pin="x:@+right"]; }
```

#edges(layout(parse(src.text).first(), layout-algo:
"tree")).first().statements.at("pin", default: none)
```

### 5.3.2.1 Parameters

```
parse(
 input: string bytes ,
 default-node-data: any ,
 eval-node-fields: string array ,
 default-edge-data: any ,
 eval-edge-fields: string array ,
 default-source-data: any ,
 eval-source-fields: string array ,
 default-sink-data: any ,
 eval-sink-fields: string array ,
 eval-graph-fields: string array ,
 eval-mode: string ,
 scope: dictionary
) -> array
```

#### 5.3.2.1.1 input string or bytes

DOT source text containing one or more graph or digraph definitions.

#### 5.3.2.1.2 default-node-data any

Default data merged into every node data. Captured node data fields override it, but bare dot statements don't set data fields. To turn statements into data fields, use `eval-node-fields`.

```
#let a = ``dot
digraph {
 a -> b -> c -> d -> a
 a -> a
 b -> d
 c [particle="q"]
}
``

#let g = parse(a.text,default-node-data:(particle:"g")).at(0)
#nodes(g).map(n=>n.data.particle)
```

Default: `none`

#### 5.3.2.1.3 eval-node-fields string or array

Node fields to evaluate into node data. The eval scope includes node statements plus node, name, and pos.

```
#let src = ``dot
digraph { a [label="#str(name)"]; }
``

#nodes(parse(src.text, eval-node-fields: "label").first()).first().data.label
```

Default: `()`

#### 5.3.2.1.4 default-edge-data any

Default data merged into every edge data. Captured edge data fields override it, but bare dot statements don't set data fields. To turn statements into data fields, use `eval-edge-fields`.

Default: `none`

#### 5.3.2.1.5 eval-edge-fields string or array

Edge statement fields to evaluate into `graph.edges(g).at(i).data`. The eval scope includes edge statements plus edge, orientation, endpoint records, placement fields, and existing edge data. Drawing callbacks later expose the edge index as `eid`.

```
#let src = ``dot
digraph { a -> b [id=0, label="#orientation"]; }
``
#edges(parse(src.text, eval-edge-fields: "label").first()).first().data.label
```

Default: `()`

#### 5.3.2.1.6 default-source-data any

Default data merged into every source half-edge data. Captured source data fields override it.

Default: `none`

#### 5.3.2.1.7 eval-source-fields string or array

Source half-edge fields to evaluate into `edge.source.data`. The eval scope includes source endpoint fields statement, port-label, compass, node, and hedge, plus the surrounding edge fields.

```
#let src = ``dot
digraph { a:left:e -> b [id=0, source="#port-label"]; }
``
#edges(parse(src.text, eval-source-fields:
"statement").first()).first().source.data.statement
```

Default: `()`

#### 5.3.2.1.8 default-sink-data any

Default data merged into every sink half-edge data. Captured sink data fields override it.

Default: `none`

#### 5.3.2.1.9 eval-sink-fields `string` or `array`

Sink half-edge fields to evaluate into `edge.sink.data`. The eval scope includes sink endpoint fields `statement`, `port-label`, `compass`, `node`, and `hedge`, plus the surrounding edge fields.

```
#let src = ``dot
digraph { a -> b:0:w [id=0, sink="#str(hedge)"]; }
```


```
#edges(parse(src.text, eval-sink-fields:
"statement").first()).first().sink.data.statement
```


```

Default: ()

5.3.2.1.10 eval-graph-fields `string` or `array`

Graph statement fields to evaluate into `graph.info(g).data`. The eval scope includes graph statements and direct graph record fields.

```
#let src = ``dot
digraph {
  title = "Strong graph";
}
```


```
#info(parse(src.text, eval-graph-fields: "title").first()).data.title
```


```

Default: ()

#### 5.3.2.1.11 eval-mode `string`

Typst eval mode used for string field values.

Default: `"markup"`

#### 5.3.2.1.12 scope `dictionary`

Additional Typst names available while evaluating field values.

Default: `(:)`

### 5.3.3 node

Create a graph node item for build.

A Typst label is the node name used by `source`, `sink`, and `pos`. The optional numeric `id` fixes the resulting graph node index. Extra named arguments are captured as node data fields, so `node(<a>, label: [A], color: red)` stores `(label: [A], color: red)`. The default draw style uses `data.label` as the visible node label when present.

### 5.3.3.1 Parameters

```
node(
 ..args: label ,
 name: none label ,
 id: none int ,
 pos: none dictionary ,
 shift: none string array dictionary ,
 statements: dictionary
) -> array
```

#### 5.3.3.1.1 **..args** label

Optional positional node name. Must be a Typst label when provided; extra named arguments become data fields.

#### 5.3.3.1.2 **name** none or label

Typst node name for references.

Default: **none**

#### 5.3.3.1.3 **id** none or int

Numeric graph node index. Must be unique and in bounds when provided.

Default: **none**

#### 5.3.3.1.4 **pos** none or dictionary

Node placement.

Default: **none**

#### 5.3.3.1.5 **shift** none or string or array or dictionary

Drawing shift stored as a statement.

Default: **none**

#### 5.3.3.1.6 **statements** dictionary

Additional flat node statements. Used by DOT; values cannot nest.

Default: ( : )

### 5.3.4 source

Create a source half-edge endpoint.

node may be a node name like <a> or a numeric node index. name gives the half-edge a Typst name; id is a numeric half-edge order/index override. Extra named arguments are captured as source data fields.

#### 5.3.4.1 Parameters

```
source(
 node: label int,
 ..args: any,
 name: none label,
 id: none int,
 statement: none string,
 compass: none string
) -> dictionary
```

##### 5.3.4.1.1 node label or int

Referenced node, either by Typst label name or numeric node index.

##### 5.3.4.1.2 ..args any

Extra named arguments become source data fields.

##### 5.3.4.1.3 name none or label

Optional half-edge name used for later references.

Default: none

##### 5.3.4.1.4 id none or int

Optional numeric half-edge index/order override.

Default: none

##### 5.3.4.1.5 statement none or string

DOT-ish flat statement used for matching/joining dangling half edges.

Default: none

##### 5.3.4.1.6 compass none or string

DOT compass point such as "n", "s", "e", or "w".

Default: none

### 5.3.5 sink

Create a sink half-edge endpoint.

node may be a node name like <a> or a numeric node index. name gives the half-edge a Typst name; id is a numeric half-edge order/index override. Extra named arguments are captured as sink data fields.

#### 5.3.5.1 Parameters

```
sink(
 node: label int,
 ..args: any,
 name: none label,
 id: none int,
 statement: none string,
 compass: none string
) -> dictionary
```

##### 5.3.5.1.1 node label or int

Referenced node, either by Typst label name or numeric node index.

##### 5.3.5.1.2 ..args any

Extra named arguments become sink data fields.

##### 5.3.5.1.3 name none or label

Optional half-edge name used for later references.

Default: none

##### 5.3.5.1.4 id none or int

Optional numeric half-edge index/order override.

Default: none

##### 5.3.5.1.5 statement none or string

DOT-ish flat statement used for matching/joining dangling half edges.

Default: none

##### 5.3.5.1.6 compass none or string

DOT compass point such as "n", "s", "e", or "w".

Default: none

### 5.3.6 edge

Create a graph edge item for build.

Positional arguments may contain one source, one sink, and optionally one Typst label used as the edge name. The numeric id chooses the edge order. Extra named arguments are captured as edge data fields, so `edge (source(<a>), sink(<b>), particle: "g")` stores `(particle: "g")`. The default draw style uses `data.label` as the visible edge label when present.

#### 5.3.6.1 Parameters

```
edge(
 ..args: any,
 name: none label,
 id: none int,
 orientation: string,
 pos: none dictionary,
 shift: none string array dictionary,
 label-pos: none string array dictionary,
 label-angle: none int float string,
 bend: none int float string,
 statements: dictionary
) -> array
```

##### 5.3.6.1.1 ..args any

Source/sink half-edges and optional edge name; extra named arguments become data fields.

##### 5.3.6.1.2 name none or label

Typst edge name.

Default: `none`

##### 5.3.6.1.3 id none or int

Numeric edge order/index override.

Default: `none`

##### 5.3.6.1.4 orientation string

Edge orientation: "default", "reversed", or "undirected".

Default: `"default"`

##### 5.3.6.1.5 pos none or dictionary

Edge placement.

Default: `none`

**5.3.6.1.6 shift** `none` or `string` or `array` or `dictionary`

Drawing shift stored as a statement.

Default: `none`

**5.3.6.1.7 label-pos** `none` or `string` or `array` or `dictionary`

Edge label position stored as a statement.

Default: `none`

**5.3.6.1.8 label-angle** `none` or `int` or `float` or `string`

Edge label angle stored as a statement.

Default: `none`

**5.3.6.1.9 bend** `none` or `int` or `float` or `string`

Edge bend stored as a statement.

Default: `none`

**5.3.6.1.10 statements** `dictionary`

Additional flat edge statements. Used by DOT; values cannot nest.

Default: `( : )`

## 5.3.7 group

Create a grouped placement coordinate.

side: "+" keeps the solved coordinate non-negative and side: "-" keeps it non-positive. Groups are layout constraints and therefore require pin placement, which is the pos default.

```
#group("right", side: "+")
```

### 5.3.7.1 Parameters

```
group(
 name: string int bool,
 side: none string
) -> dictionary
```

#### 5.3.7.1.1 name `string` or `int` or `bool`

Group identifier shared by positions constrained to the same coordinate.

#### 5.3.7.1.2 side `none` or `string`

Optional sign constraint: "+", "-", "positive", or "negative".

Default: `none`

### 5.3.8 pin

Mark one coordinate as a layout constraint.

#### 5.3.8.1 Parameters

```
pin(value: int | float | dictionary) -> dictionary
```

##### 5.3.8.1.1 value `int` or `float` or `dictionary`

Coordinate value to constrain.

### 5.3.9 start

Mark one coordinate as an initial layout value only.

#### 5.3.9.1 Parameters

```
start(value: int | float) -> dictionary
```

##### 5.3.9.1.1 value `int` or `float`

Coordinate value to use as the layout seed.

### 5.3.10 pos

Create a first-class graph placement.

The default mode: "pin" turns numeric coordinates into layout constraints and also makes the coordinates immediately drawable without a layout pass. Use `start(value)` for an individual coordinate that should only seed the layout, or `pin(value)` to pin an individual numeric coordinate when mode: "start" is used. Grouped coordinates are always layout constraints for their axis.

```
#pos(x: group("right", side: "+"), y: start(10))
```

### 5.3.10.1 Parameters

```
pos(
 x: none int float dictionary ,
 y: none int float dictionary ,
 ref: none label int ,
 dx: none int float ,
 dy: none int float ,
 mode: string
) -> dictionary
```

#### 5.3.10.1.1 x none or int or float or dictionary

Absolute or grouped x coordinate.

Default: none

#### 5.3.10.1.2 y none or int or float or dictionary

Absolute or grouped y coordinate.

Default: none

#### 5.3.10.1.3 ref none or label or int

Node reference for relative placement, by name or numeric index.

Default: none

#### 5.3.10.1.4 dx none or int or float

Relative x offset from ref.

Default: none

#### 5.3.10.1.5 dy none or int or float

Relative y offset from ref.

Default: none

#### 5.3.10.1.6 mode string

Placement mode: "pin" constrains layout, "start" only seeds it.

Default: "pin"

### 5.3.11 map

Map graph metadata to new native data.

The callbacks receive decoded records plus a fields dictionary containing merged statements and direct record fields. A callback returns `none` to leave the record unchanged, `(data: value)` to set new native data, or structural fields such as `pos`, `shift`, and `statements` to patch data seen by later layout calls.

```
#let g = build({ node(<a>) })
#let g = map(g, node: node => (data: (label: [A])))
#nodes(g).first().data.label
```

#### 5.3.11.1 Parameters

```
map(
 graph_: dictionary ,
 graph: none function ,
 node: none function ,
 edge: none function ,
 source: none function ,
 sink: none function
) -> dictionary
```

##### 5.3.11.1.1 graph\_ dictionary

Graph object to transform.

##### 5.3.11.1.2 graph none or function

Callback for graph metadata records.

Default: `none`

##### 5.3.11.1.3 node none or function

Callback for node records.

Default: `none`

##### 5.3.11.1.4 edge none or function

Callback for edge records.

Default: `none`

##### 5.3.11.1.5 source none or function

Callback for source half-edge records.

Default: `none`

#### 5.3.11.1.6 sink `none` or `function`

Callback for sink half-edge records.

Default: `none`

### 5.3.12 style

Attach layout-relevant drawing style to a graph.

Node style is measured immediately and stored as `layout-width` / `layout-height` statements for later layout calls. Edge labels are measured as `label-width` / `label-height` statements for label placement. `draw` uses the stored node and edge-label style by default.

#### 5.3.12.1 Parameters

```
style(
 graph_: dictionary ,
 scope: dictionary ,
 unit: int float length ratio ,
 node-label: auto content string function none ,
 node-label-style: dictionary function ,
 node-style: dictionary function none ,
 edge-label: content string function none ,
 edge-label-style: dictionary function
) -> dictionary
```

##### 5.3.12.1.1 graph\_ dictionary

Graph object to style.

##### 5.3.12.1.2 scope dictionary

Extra scope visible to label/style callbacks.

Default: `(:)`

##### 5.3.12.1.3 unit int or float or length or ratio

Coordinate length for one graph-layout unit. Numbers are interpreted as em.

Default: `1`

##### 5.3.12.1.4 node-label auto or content or string or function or none

Node label content or callback. `auto` uses node data/statement label or node name.

Default: `auto`

#### 5.3.12.15 node-label-style    dictionary or function

CeTZ content style for node labels. Its padding contributes to measured node size.

Default: ( : )

#### 5.3.12.16 node-style    dictionary or function or none

CeTZ node shape style or callback. Explicit numeric radii contribute to measured node size.

Default: ( : )

#### 5.3.12.17 edge-label    content or string or function or none

Edge label content or callback. none leaves edge labels unstyled and unmeasured.

Default: none

#### 5.3.12.18 edge-label-style    dictionary or function

CeTZ content style for edge labels. Its padding contributes to measured edge-label size.

Default: ( : )

### 5.3.13 eval-fields

Evaluate selected fields into native data entries.

Each selected field is read from the record's merged fields dictionary, evaluated in a scope containing those fields, and written to `data.<field>`.

```
#let g = build({
 node(<a>)
 node()
 edge(source(<a>), sink(), statements: (mom: "p"))
}, default-edge-statements: (display-label: "$#mom$"))
#let g = eval-fields(g, eval-edge-fields: ("display-label",))
#edges(g).first().data.at("display-label")
```

### 5.3.13.1 Parameters

```
eval-fields(
 graph_: dictionary ,
 eval-graph-fields: string array ,
 eval-node-fields: string array ,
 eval-edge-fields: string array ,
 eval-source-fields: string array ,
 eval-sink-fields: string array ,
 eval-mode: string ,
 scope: dictionary
) -> dictionary
```

#### 5.3.13.1.1 graph\_ dictionary

Graph object whose selected statement fields should be evaluated.

#### 5.3.13.1.2 eval-graph-fields string or array

Graph statement fields to evaluate into `graph.info(g).data`.

Default: ()

#### 5.3.13.1.3 eval-node-fields string or array

Node statement fields to evaluate into node data.

Default: ()

#### 5.3.13.1.4 eval-edge-fields string or array

Edge statement fields to evaluate into edge data.

Default: ()

#### 5.3.13.1.5 eval-source-fields string or array

Source half-edge fields to evaluate into source data.

Default: ()

#### 5.3.13.1.6 eval-sink-fields string or array

Sink half-edge fields to evaluate into sink data.

Default: ()

#### 5.3.13.1.7 eval-mode string

Typst eval mode used for string field values.

Default: "markup"

#### 5.3.13.1.8 scope dictionary

Additional Typst names available while evaluating field values.

Default: ( : )

### 5.3.14 info

Return graph metadata.

The result has name, global-statements, default-edge-statements, and default-node-statements.

```
#let g = build({ node(<a>) }, name: "demo")
#info(g).name
```

#### 5.3.14.1 Parameters

`info(graph: dictionary) -> dictionary`

#### 5.3.14.1.1 graph dictionary

Graph object returned by build, parse, layout, or another graph API.

### 5.3.15 dot

Serialize a graph object to DOT.

```
#let g = build({
 node(<a>)
 node()
 edge(source(<a>), sink())
}, name: "demo")
#dot(g).contains("digraph demo")
```

#### 5.3.15.1 Parameters

`dot(graph: dictionary) -> string`

#### 5.3.15.1.1 graph dictionary

Graph object to serialize.

### 5.3.16 nodes

Return node records, optionally filtered by a subgraph object.

Node name values are Typst labels when present.

```
#let g = build({
 node(<a>)
 node()
})
#nodes(g).map(node => str(node.name)).join(", ")
```

#### 5.3.16.1 Parameters

```
nodes(
 graph: dictionary,
 subgraph: none bytes
) -> array
```

#### 5.3.16.1.1 graph dictionary

Graph object to inspect.

#### 5.3.16.1.2 subgraph none or bytes

Optional subgraph filter; only nodes incident to selected half edges are returned.

Default: none

### 5.3.17 edges

Return edge records, optionally filtered by a subgraph object.

Edge name values are Typst labels when present.

```
#let g = build({
 node(<a>)
 node()
 edge(source(<a>, compass: "e"), sink())
})
#edges(g).len()
```

### 5.3.17.1 Parameters

```
edges(
 graph: dictionary,
 subgraph: none bytes
) -> array
```

#### 5.3.17.1.1 graph dictionary

Graph object to inspect.

#### 5.3.17.1.2 subgraph none or bytes

Optional subgraph filter; only selected edges/half-edges are returned.

Default: none

### 5.3.18 node-data

Return one named node's native data.

name is a Typst label such as <a> or the corresponding string name.

```
#let g = build({ node(<a>, label: [A]) })
#node-data(g, <a>).label
```

### 5.3.18.1 Parameters

```
node-data(
 graph_: dictionary,
 name: label string
) -> any
```

#### 5.3.18.1.1 graph\_ dictionary

Graph object to inspect.

#### 5.3.18.1.2 name label or string

Node name as a Typst label or its string form.

### 5.3.19 edge-data

Return one named edge's native data.

name is a Typst label such as <e> or the corresponding string name.

```
#let g = build({
 node(<a>)
 node()
 edge(source(<a>), <e>, sink(), label: [p])
})
#edge-data(g, <e>).label
```

#### 5.3.19.1 Parameters

```
edge-data(
 graph_: dictionary,
 name: label string
) -> any
```

##### 5.3.19.1.1 graph\_ dictionary

Graph object to inspect.

##### 5.3.19.1.2 name label or string

Edge name as a Typst label or its string form.

#### 5.3.20 update-node-data

Update one named node's native data.

update may be a replacement data value or a function (data, node) => new-data.

```
#let g = build({ node(<a>) })
#let g = update-node-data(g, <a>, (label: [A]))
#nodes(g).first().data.label
```

#### 5.3.20.1 Parameters

```
update-node-data(
 graph_: dictionary,
 name: label string,
 update: any function
) -> dictionary
```

##### 5.3.20.1.1 graph\_ dictionary

Graph object to update.

##### 5.3.20.1.2 name label or string

Node name as a Typst label or its string form.

#### 5.3.20.1.3 update any or function

Replacement data or (data, node) => new-data callback.

### 5.3.21 update-edge-data

Update one named edge's native data.

update may be a replacement data value or a function (data, edge) => new-data.

```
#let g = build({
 node(<a>)
 node()
 edge(source(<a>), <e>, sink())
})
#let g = update-edge-data(g, <e>, (label: [p]))
#edges(g).first().data.label
```

#### 5.3.21.1 Parameters

```
update-edge-data(
 graph_: dictionary,
 name: label string,
 update: any function
) -> dictionary
```

##### 5.3.21.1.1 graph\_ dictionary

Graph object to update.

##### 5.3.21.1.2 name label or string

Edge name as a Typst label or its string form.

##### 5.3.21.1.3 update any or function

Replacement data or (data, edge) => new-data callback.

### 5.3.22 join

Join two graphs by matching dangling half-edge statements or ids on key.

Supported key values are "statement", "compass", and "id".

```
#let left = build({
 node(<a>)
 edge(sink(<a>, statement: "j"))
})
#let right = build({
 node()
 edge(source(, statement: "j"))
})
#edges(join(left, right, key: "statement")).len()
```

### 5.3.22.1 Parameters

```
join(
 left: dictionary,
 right: dictionary,
 key: string
) -> dictionary
```

#### 5.3.22.1.1 left dictionary

Left graph object.

#### 5.3.22.1.2 right dictionary

Right graph object.

#### 5.3.22.1.3 key string

Dangling half-edge match key: "statement", "compass", or "id".

Default: "statement"

### 5.3.23 cycles

Return subgraph objects for the graph's cycle basis.

```
#let g = build({
 node(<a>)
 node()
 edge(source(<a>), sink())
})
#cycles(g).len()
```

#### 5.3.23.1 Parameters

```
cycles(graph: dictionary) -> array
```

#### 5.3.23.1.1 graph dictionary

Graph object to analyze.

### 5.3.24 forests

Return subgraph objects for the graph's spanning forests.

```
#let g = build({
 node(<a>)
 node()
 edge(source(<a>), sink())
})
#forests(g).len()
```

#### 5.3.24.1 Parameters

`forests`(graph: dictionary) -> array

#### 5.3.24.1.1 graph dictionary

Graph object to analyze.

## 5.4 layout

- `layout()`
- `sequence()`

### 5.4.1 layout

Apply the linest layout pass to a graph object.

This is intentionally a second step: construct or parse a graph first, then call `layout`. Set `layout-algo` to "force" for deterministic force integration, "anneal" for simulated annealing, "tree" for a traversal tree placement, "dot" for a Graphviz-like layered placement, or "stable-layered" for a stable railroad-inspired layered placement.

```
#let g = graph.parse("digraph partial { a -> b; a -> b; a:s -> b:s; b:s -> c:s; c:s -> d:s; d:s -> a:s }").at(0)
#let south = subgraph.compass(g, "s")
#let gf = layout(layout(g, layout-algo: "force"), layout-algo: "tree", layout-nodes: "fixed", subgraph:south)
#let ga = layout(g, layout-algo: "force")
#let gt = layout(layout(g, layout-algo: "tree", layout-roots: (2)), layout-algo: "anneal", layout-nodes: "fixed", gamma-ee:0.1, gamma-ev:.75, beta:5, length-scale:0.4)
#grid(columns: 3, gutter: 2cm, draw(gf), draw(ga), draw(gt))
```

```
#let g = graph.parse("digraph partial { a -> b; b -> c; c -> d; d -> a }").at(0)
#let tree = graph.forests(g).at(0)
#let g = layout(g, layout-algo: "tree", subgraph: tree)
#graph.edges(g).map(edge => edge.pos)
```

### 5.4.1.1 Parameters

```
layout(
 graph: dictionary ,
 subgraph: none bytes ,
 viewport-w: float ,
 viewport-h: float ,
 tree-dx: float ,
 tree-dy: float ,
 steps: int ,
 seed: int ,
 step: float ,
 step-shrink: float ,
 cool: float ,
 accept-floor: float ,
 early-tol: float ,
 temp: float ,
 delta: float ,
 beta: float ,
 k-spring: float ,
 g-center: float ,
 epochs: int ,
 crossing-penalty: float ,
 gamma-dangling: float ,
 gamma-ee: float ,
 directional-force: float ,
 label-length-scale: float ,
 label-spring: float ,
 label-charge: float ,
 label-steps: int ,
 label-layout: string ,
 label-step: float ,
 label-early-tol: float ,
 label-max-delta-scale: float ,
 gamma-ev: float ,
 eps: float ,
 incremental-energy: bool ,
 layout-algo: string ,
 layout-nodes: string ,
 layout-direction: string ,
 rank-align: string ,
 layout-roots: array ,
 rank-same: array ,
 route-edge-weight: float ,
 route-exit-weight: float ,
 route-label-width-scale: float ,
 route-label-width-cap: float ,
 z-spring: float ,
 z-spring-growth: float ,
 length-scale: float
) -> dictionary
```

#### 5.4.1.1.1 graph dictionary

Graph object returned by `graph.build` or `graph.parse`.

#### 5.4.1.1.2 subgraph `none` or bytes

Optional subgraph object to lay out. With "tree", other edges are drawn from the resulting node positions. With "dot" and "stable-layered", the subgraph determines rank constraints, while all paired edges between included nodes get dummy routing vertices and edge positions. With "force" and "anneal", nodes and edges outside the subgraph are fixed boundary points during optimization.

Default: `none`

#### 5.4.1.1.3 viewport-w `float`

Width of the layout viewport used to derive the natural spring length. Applies to both "force" and "anneal".

Default: `10.0`

#### 5.4.1.1.4 viewport-h `float`

Height of the layout viewport used to derive the natural spring length. Applies to both "force" and "anneal".

Default: `10.0`

#### 5.4.1.1.5 tree-dx `float`

Horizontal spacing multiplier for traversal-tree and layered placement. For "force" and "anneal", this scales the initial placement.

Default: `0.9`

#### 5.4.1.1.6 tree-dy `float`

Vertical spacing multiplier for traversal-tree and layered placement. For "force" and "anneal", this scales the initial placement.

Default: `1.2`

#### 5.4.1.1.7 steps `int`

Iterations per epoch. In "force" mode this is the number of force integration steps; in "anneal" mode this is the number of proposals per temperature epoch.

Default: `int(sys.inputs.at("steps", default: "30"))`

#### 5.4.1.1.8 seed `int`

Seed for deterministic initialization, force-mode jitter, and annealing proposals. Applies to both modes.

Default: `int(sys.inputs.at("seed", default: "2"))`

#### 5.4.1.1.9 **step** float

Initial movement scale. "force" multiplies computed forces by this value; "anneal" uses it as a proposal step size in natural spring-length units.

Default: 0.81

#### 5.4.1.1.10 **step-shrink** float

Anneal-only step shrink factor, applied when an epoch's acceptance ratio falls below accept-floor.

Default: 0.21

#### 5.4.1.1.11 **cool** float

Cooling factor applied once per epoch. "anneal" cools temp; "force" shrinks step.

Default: 0.85

#### 5.4.1.1.12 **accept-floor** float

Anneal-only acceptance-ratio threshold below which step is shrunk by step-shrink.

Default: 0.15

#### 5.4.1.1.13 **early-tol** float

Force-only early stop threshold for maximum movement in one step, as a multiple of the natural spring length. The annealing schedule stores this value but does not currently use it for stopping.

Default: 1e-6

#### 5.4.1.1.14 **temp** float

Anneal-only initial temperature used in the Metropolis acceptance test, scaled by natural spring length squared.

Default: 0.3

#### 5.4.1.1.15 **delta** float

Force-mode maximum movement clamp per point and per step, as a multiple of the natural spring length.

Default: 0.4

#### 5.4.1.1.16 **beta** float

Base repulsion strength for vertex-vertex interactions. Also scales gamma-ev, gamma-ee, gamma-dangling, and g-center. Applies to both modes through the shared spring energy.

Default: 50.0

#### 5.4.1.1.17 **k-spring** float

Spring stiffness for node-to-edge incidence lengths. Applies to both modes.

Default: 11.0

#### 5.4.1.1.18 **g-center** float

Centering strength relative to beta. Applies to both modes.

Default: 0.002

#### 5.4.1.1.19 **epochs** int

Number of epochs. Both modes run up to steps iterations inside each epoch.

Default: 30

#### 5.4.1.1.20 **crossing-penalty** float

Anneal-only fixed energy penalty per detected edge crossing. The direct force integrator does not currently add a crossing force.

Default: 30.0

#### 5.4.1.1.21 **gamma-dangling** float

Repulsion for dangling half edges, relative to beta. Applies to both modes through the shared spring energy.

Default: 5.0

#### 5.4.1.1.22 **gamma-ee** float

Local edge-edge repulsion, relative to beta. Applies to both modes.

Default: 0.1

#### 5.4.1.1.23 directional-force float

Bias that pushes points in directions implied by pin/port constraints. Force mode treats this as a force and scales it by natural spring length; anneal mode treats it as a dimensionless multiplier on proposal steps. Applies to both modes.

Default: 5.0

#### 5.4.1.1.24 label-length-scale float

Edge-label target offset as a multiple of the graph spring length. Label layout runs after both graph layout modes.

Default: 0.6

#### 5.4.1.1.25 label-spring float

Spring strength pulling each label toward its target offset in the spring-based label layouts.

Default: 23.0

#### 5.4.1.1.26 label-charge float

Repulsion strength between labels and graph points, scaled by spring length squared. Label layout runs after both modes.

Default: 3.0

#### 5.4.1.1.27 label-steps int

Maximum number of post-layout label relaxation steps. Set to 0 to skip label placement. Applies after both modes.

Default: 20

#### 5.4.1.1.28 label-layout string

Edge-label relaxation model. "normal" uses a perpendicular offset, "dangling-tangent" uses the edge direction for dangling half-edge labels and a perpendicular offset for paired edges, and "fixed-length" keeps each label at a fixed distance from its edge point and only lets that segment rotate.

Default: "normal"

#### 5.4.1.1.29 label-step float

Label relaxation step size. Applies after both modes.

Default: 0.15

#### 5.4.1.1.30 label-early-tol float

Label relaxation early stop threshold. Applies after both modes.

Default: `1e-3`

#### 5.4.1.1.31 label-max-delta-scale float

Label movement clamp as a multiple of spring length. Applies after both modes.

Default: `0.5`

#### 5.4.1.1.32 gamma-ev float

Edge-vertex repulsion, relative to beta. Applies to both modes.

Default: `0.01`

#### 5.4.1.1.33 eps float

Softening epsilon used in inverse-square force/energy terms. Applies to both modes.

Default: `1e-4`

#### 5.4.1.1.34 incremental-energy bool

Whether annealing updates cached energy by local deltas. This is anneal-only; force mode computes direct forces instead of energies.

Default: `true`

#### 5.4.1.1.35 layout-algo string

Layout algorithm. Use "force" for direct force integration, "anneal" for simulated annealing against the spring energy, "tree" for a traversal-tree placement, "dot" for a Graphviz-like layered placement, or "stable-layered" for a stable railroad-inspired layered placement.

Default: `"force"`

#### 5.4.1.1.36 layout-nodes string

Node movement policy. "layout" lets the layout algorithm move nodes. "fixed" keeps every node at its current position for this layout pass and only moves edge control points. With subgraph, only edges in the subgraph are moved; other edge control points stay at their current positions.

Default: `"layout"`

#### 5.4.1.1.37 layout-direction string

Direction for traversal-tree and layered rank placement. "down" places increasing ranks downward; "right" swaps the layout axes so increasing ranks go left-to-right and measured node/label widths reserve rank-axis space.

Default: "down"

#### 5.4.1.1.38 rank-align string

Alignment of real nodes inside a rank along the rank axis. "center" keeps node centers aligned, while "start" / "left" and "end" / "right" align the corresponding measured node-box side.

Default: "center"

#### 5.4.1.1.39 layout-roots array

Ordered node indices used as preferred roots for "tree", "dot", and "stable-layered". Roots outside the selected node set are ignored. Remaining components are laid out afterward in graph order.

Default: ()

#### 5.4.1.1.40 rank-same array

Subgraphs whose incident nodes should share a dot/stable-layered rank. These are layout hints supplied by Typst rather than parsed graph structure.

Default: ()

#### 5.4.1.1.41 route-edge-weight float

Dot/stable-layered also honors a node statement layout-rank as an exact non-negative integer rank. Nodes with the same layout-rank are placed on the same horizontal layer, and larger ranks are placed lower. Relative layout weight for paired edges outside the dot/stable-layered rank subgraph. Lower values make these edges guide routing without dominating the rank tree.

Default: 0.15

#### 5.4.1.1.42 route-exit-weight float

Extra horizontal straightening weight for the first or last segment of an edge with source-route-exit or sink-route-exit set to a vertical side.

Default: 4.0

#### 5.4.1.143 route-label-width-scale float

Multiplier for measured edge-label width when sizing non-rank dummy routing vertices in dot/stable-layered layout.

Default: 1.0

#### 5.4.1.144 route-label-width-cap float

Maximum non-rank dummy label width as a multiple of `tree-dx`. Set to 0 or a negative value to disable the cap.

Default: 2.0

#### 5.4.1.145 z-spring float

Force-only spring pulling temporary z coordinates back toward the layout plane. Higher values keep the visible 2D forces from being hidden by the temporary 3D symmetry-breaking offsets.

Default: 2.0

#### 5.4.1.146 z-spring-growth float

Force-only per-epoch multiplier for z-spring.

Default: 1.0

#### 5.4.1.147 length-scale float

Natural spring-length multiplier. This scales the graph's preferred edge length and dimensional force/energy terms so changing only this value mostly zooms the result instead of retuning the force ratios. Applies to both modes.

Default: 0.35

### 5.4.2 sequence

Apply multiple layout passes in order.

Each pass is a dictionary of named arguments accepted by `layout`, excluding the graph itself.

#### 5.4.2.1 Parameters

```
sequence(
 graph: dictionary ,
 passes: array
) -> dictionary
```

#### 5.4.2.1.1 **graph**      dictionary

Graph object returned by `graph.build` or `graph.parse`.

#### 5.4.2.1.2 **passes**      array

Array of layout option dictionaries.

## 5.5 **physics**

- `stroke-style()`
- `source-stroke()`
- `sink-stroke()`
- `particle-name()`
- `edge-entry()`
- `text-value()`
- `eval-expression()`
- `label-content()`
- `style-dict()`
- `momentum-value()`
- `edge-index()`
- `dangling-half-edge-index()`
- `source-style()`
- `sink-style()`
- `edge-label()`
- `style()`

### **Variables**

- `massive`
- `massless`
- `dashed`
- `dotted`
- `fermion-arrow-mark`
- `fermion-flow`
- `wave`
- `coil`
- `zigzag`
- `default-edge`
- `default-map`
- `momentum-arrow-defaults`

### 5.5.1 **stroke-style**

Build a rounded CeTZ stroke dictionary accepted by `linnest.draw`.

#### 5.5.1.1 **Parameters**

```
stroke-style(
 c,
 thickness,
 dash
) -> dictionary
```

### 5.5.2 source-stroke

Source-half stroke helper.

#### 5.5.2.1 Parameters

```
source-stroke(
 c,
 thickness,
 dash
) -> dictionary
```

### 5.5.3 sink-stroke

Sink-half stroke helper. The default lightening makes the two halves encode the graph's source/sink split without requiring arrowheads.

#### 5.5.3.1 Parameters

```
sink-stroke(
 c,
 thickness,
 dash,
 lighten
) -> dictionary
```

### 5.5.4 particle-name

Return an edge's particle name, stripping the quotes often present in DOT statement values.

#### 5.5.4.1 Parameters

```
particle-name(edge) -> none string
```

### 5.5.5 edge-entry

Return the style-map entry for an edge.

#### 5.5.5.1 Parameters

```
edge-entry(
 edge,
 map,
 default
) -> dictionary
```

### 5.5.6 text-value

Convert DOT-ish values to plain text.

#### 5.5.6.1 Parameters

```
text-value(value) -> string
```

#### 5.5.7 eval-expression

Evaluate a string-valued style expression in the edge scope.

##### 5.5.7.1 Parameters

```
eval-expression(
 value,
 edge,
 map,
 scope
) -> any
```

#### 5.5.8 label-content

Convert a style label value to content. In mode: "eval", string values are evaluated in the edge scope.

##### 5.5.8.1 Parameters

```
label-content(
 value,
 edge,
 mode,
 map,
 scope
) -> none content
```

#### 5.5.9 style-dict

Convert a style value to a dictionary. In mode: "eval", string values are evaluated in the edge scope.

##### 5.5.9.1 Parameters

```
style-dict(
 value,
 edge,
 mode,
 map,
 scope
) -> dictionary
```

#### 5.5.10 momentum-value

Return the momentum field used by optional edge labels.

#### 5.5.10.1 Parameters

```
momentum-value(
 edge,
 fields
) -> none any
```

#### 5.5.11 edge-index

Return the edge index used by optional edge labels. The DOT id statement wins over the renderer-local eid.

##### 5.5.11.1 Parameters

```
edge-index(
 edge,
 fields
) -> none any
```

#### 5.5.12 dangling-half-edge-index

Return the exposed half-edge index for dangling edges only.

##### 5.5.12.1 Parameters

```
dangling-half-edge-index(edge) -> none any
```

#### 5.5.13 source-style

Style callback for the source half edge.

momentum-arrows: true adds one centered black CeTZ-mark decoration while the main edge remains drawn with its normal particle style.

##### 5.5.13.1 Parameters

```
source-style(
 edge,
 map,
 default,
 typst-fields,
 scope,
 orientation-split,
 momentum-arrows,
 momentum-arrow-offset,
 momentum-arrow-length,
 momentum-arrow-ratio,
 momentum-arrow-stroke,
 momentum-arrow-mark
) -> dictionary array
```

#### 5.5.14 sink-style

Style callback for the sink half edge.

`momentum-arrows: true` adds one centered black CeTZ-mark decoration toward the sink node, so momentum arrows always flow from source to sink independently of `edge.orientation`.

##### 5.5.14.1 Parameters

```
sink-style(
 edge,
 map,
 default,
 typst-fields,
 scope,
 orientation-split,
 momentum-arrows,
 momentum-arrow-offset,
 momentum-arrow-length,
 momentum-arrow-ratio,
 momentum-arrow-stroke,
 momentum-arrow-mark
) -> dictionary array
```

#### 5.5.15 edge-label

Edge-label callback.

By default this preserves data `display-label` or `label`, then explicit `display-label`, `label`, and `particle-map` labels. Set any `show-*` option to build a label from selected metadata instead:

```
#let callbacks = physics.style(
 momentum-arrows: true,
 show-edge-index: true,
 show-particle: true,
)
```

### 5.5.15.1 Parameters

```
edge-label(
 edge,
 map,
 default,
 typst-fields,
 scope,
 show-momentum,
 show-edge-index,
 show-half-edge-index,
 show-particle,
 momentum-fields,
 edge-index-fields,
 momentum-prefix,
 edge-index-prefix,
 half-edge-index-prefix,
 particle-prefix,
 label-separator,
],
 label-size,
 label-fill
) -> none content
```

### 5.5.16 style

Bundle source-style, sink-style, and edge-label callbacks with shared options for `linnest.draw`.

```
#let g = build({
 node(<a>, pos: pos(y:0))
 node(, pos: pos(y:0))
 edge(source(<a>), <g-edge>, sink(), particle: "g")
 edge(source(<a>), <g-edge2>, sink(), particle: "g")
 edge(source(<a>), <g-edge3>, sink(), particle: "g")
},
 name: "physics",
)
#let a = ``dot
digraph {
 0 [dod="-100" num="1"];
 1 [dod="-100" num="1"];
 2 [dod="-100" num="1"];
 3 [dod="-100" num="1"];
 4 [dod="-100" num="1"];
 5 [dod="-100" num="1"];
 6 [dod="-100" num="1"];
 exte0 [style=invis];
 exte0 -> 2:0 [id=0 dod="-100" lmb_rep="P(0,a___)" num="1" particle="d" pin="x:@-
left"];
 exte1 [style=invis];
 exte1 -> 1:1 [id=1 dir=back dod="-100" lmb_rep="P(1,a___)" num="1" particle="d~"
pin="x:@-left"];
 exte2 [style=invis];
 exte2 -> 1:2 [id=2 dir=none dod="-100" lmb_rep="P(2,a___)" mass="0" num="1"
particle="H" pin="x:@-left"];
```

[illegible]

### 5.5.16.1 Parameters

```
style(
 map,
 default,
 typst-fields,
 scope,
 orientation-split,
 momentum-arrows,
 momentum-arrow-offset,
 momentum-arrow-length,
 momentum-arrow-ratio,
 momentum-arrow-stroke,
 momentum-arrow-mark,
 show-momentum,
 show-edge-index,
 show-half-edge-index,
 show-particle,
 momentum-fields,
 edge-index-fields,
 momentum-prefix,
 edge-index-prefix,
 half-edge-index-prefix,
 particle-prefix,
 label-separator,
],
 label-size,
 label-fill
) -> dictionary
```

length

### 5.5.17 massive

Conventional massive-particle stroke thickness.

length

### 5.5.18 massless

Conventional massless-particle stroke thickness.

array

### 5.5.19 dashed

Dash pattern used for scalar-style lines.

string

### 5.5.20 dotted

Dotted dash style used for ghost-style lines.

dictionary

#### 5.5.21 fermion-arrow-mark

CeTZ marker used for fermion particle-flow arrows on the main edge.

dictionary

#### 5.5.22 fermion-flow

Mark an edge-map entry as a fermion so the main edge receives one particle-flow arrow that follows `edge.orientation`. This is separate from optional momentum arrows.

dictionary

#### 5.5.23 wave

Photon-style wave pattern.

dictionary

#### 5.5.24 coil

Gluon-style coil pattern.

dictionary

#### 5.5.25 zigzag

Weak-boson-style zigzag pattern.

dictionary

#### 5.5.26 default-edge

Fallback style entry used when an edge has no known particle type.

dictionary

#### 5.5.27 default-map

Small built-in particle map for standalone physics diagrams. GammaLoop generated `edge-style.typ` files pass their model-specific map to `style`.

dictionary

#### 5.5.28 momentum-arrow-defaults

Default style for source-to-sink momentum arrow layers.

### 5.6 draw

- `edge-halves()`

- `to-cetz-edge-halves()`
- `draw()`

### 5.6.1 edge-halves

Split a laid-out graph edge into source and sink half-edge paths.

The returned dictionary has source, sink, and curve. The split point is the edge layout point, so the two half-edges join smoothly there.

#### 5.6.1.1 Parameters

```
edge-halves(
 edge: dictionary,
 nodes: array,
 omega: float,
 source-outset: int float,
 sink-outset: int float,
 accuracy: float
) -> dictionary
```

##### 5.6.1.1.1 edge dictionary

Edge record returned by `graph.edges(layout(g))`.

##### 5.6.1.1.2 nodes array

Node records returned by `graph.nodes(layout(g))`.

##### 5.6.1.1.3 omega float

Hobby curl used for the source-to-edge and edge-to-sink curves.

Default: 1.0

##### 5.6.1.1.4 source-outset int or float

Arc-length trim applied at the source node side.

Default: 0

##### 5.6.1.1.5 sink-outset int or float

Arc-length trim applied at the sink node side.

Default: 0

#### 5.6.1.1.6 accuracy float

Arc-length accuracy used while trimming.

Default: 0.001

### 5.6.2 to-cetz-edge-halves

Draw the two halves of a laid-out graph edge through CeTZ.

source-style applies from the source node to the edge layout point, and sink-style applies from the edge layout point to the sink node.

#### 5.6.2.1 Parameters

```
to-cetz-edge-halves(
 edge: dictionary ,
 nodes: array ,
 unit: int float length ratio ,
 omega: float ,
 source-outset: int float ,
 sink-outset: int float ,
 accuracy: float ,
 source-style: dictionary ,
 sink-style: dictionary
) -> content
```

##### 5.6.2.1.1 edge dictionary

Edge record returned by `graph.edges(layout(g))`.

##### 5.6.2.1.2 nodes array

Node records returned by `graph.nodes(layout(g))`.

##### 5.6.2.1.3 unit int or float or length or ratio

Coordinate length for one graph-layout unit. Numbers are interpreted as em.

Default: 1

##### 5.6.2.1.4 omega float

Hobby curl used for the source-to-edge and edge-to-sink curves.

Default: 1.0

#### 5.6.2.1.5 source-outset `int` or `float`

Arc-length trim applied at the source node side.

Default: `0`

#### 5.6.2.1.6 sink-outset `int` or `float`

Arc-length trim applied at the sink node side.

Default: `0`

#### 5.6.2.1.7 accuracy `float`

Arc-length accuracy used while trimming.

Default: `0.001`

#### 5.6.2.1.8 source-style `dictionary`

CeTZ style for the source half edge.

Default: `(:)`

#### 5.6.2.1.9 sink-style `dictionary`

CeTZ style for the sink half edge.

Default: `(:)`

### 5.6.3 draw

Draw a graph object with CeTZ.

The graph must already have positions, either from layout or from explicit pos values passed to graph node or edge items. Paired edges without an explicit edge position use the midpoint of their endpoint nodes. Node and edge Typst style dictionaries or callbacks are evaluated and forwarded to CeTZ.

```
#let positioned = graph.build({
 graph.node(<left>, label: [left], pos: graph.pos(x: 0, y: 0))
 graph.node(<right>, label: [right], pos: graph.pos(ref: <left>, dx: 2.5, dy: 0))
 graph.edge(graph.source(<left>), graph.sink(<right>))
 graph.edge(graph.source(<right>), <right-out>, pos: graph.pos(ref: <right>, dx: 0.9, dy: 0.7))
},
 name: "positioned",
)
#draw(positioned, source-style: (stroke: black + 0.7pt), sink-style: (stroke: black + 0.7pt))

#let parallel-base-edge = 0
```

```

#let source-patterns = (
 "coil",
 "zigzag",
 "coil",
 "wave",
 "zigzag",
 "coil",
 "wave",
 "zigzag",
)
#let sink-patterns = (
 "coil",
 "zigzag",
 "coil",
 "zigzag",
 "coil",
 "wave",
 "coil",
 "wave",
)
#let parallel-layer(edge, mark: none) = if edge.eid == parallel-base-edge {
 (
 offset: -0.5,
 length: 1.5,
 ratio: 0.5,
 resolve-length: "min",
 stroke: (paint: rgb("#2f6f4e"), thickness: 1.1pt, cap: "round"),
 mark: mark,
)
} else { none }
#let stack(base, layer) = if layer == none { base } else { (base, layer) }
#let source-stroke(edge) = if edge.eid == parallel-base-edge {
 (paint: gray, thickness: 0.75pt, cap: "round")
} else {
 (paint: red, thickness: 1.2pt, cap: "round")
}
#let sink-stroke(edge) = if edge.eid == parallel-base-edge {
 (paint: gray, thickness: 0.75pt, cap: "round")
} else {
 (paint: blue, thickness: 1.2pt, cap: "round", dash: "dotted")
}
#let source-style(edge) = {
 let base = (
 stroke: source-stroke(edge),
 pattern: source-patterns.at(edge.eid),
 pattern-amplitude: 0.18,
 pattern-wavelength: 0.55,
 pattern-coil-longitudinal-scale: 1.6,
)
 stack(base, parallel-layer(edge))
}
#let sink-style(edge) = {
 let base = (
 stroke: sink-stroke(edge),
 pattern: sink-patterns.at(edge.eid),
 pattern-amplitude: 0.18,
 pattern-wavelength: 0.55,
 pattern-coil-longitudinal-scale: 1.6,
)
 stack(base, parallel-layer(edge, mark: (end: (symbol: "straight"), scale: 0.75)))
}
#let g = graph.build({

```

```

graph.node(<a>, label: [a hi])
graph.node(<c>)
graph.node(<d>)
graph.node(<e>)
graph.edge(graph.source(<a>), <ac>, graph.sink(<c>), pos: graph.pos(x: 0, y: 0.75, mode: "pin"))
graph.edge(graph.source(<c>), <ca>, graph.sink(<a>, compass: "e"))
graph.edge(graph.source(<c>), <cd>, graph.sink(<d>, compass: "e"))
graph.edge(graph.source(<e>), <ed>, graph.sink(<d>, compass: "e"))
graph.edge(graph.source(<e>), <ea>, graph.sink(<a>, compass: "e"))
graph.edge(graph.source(<d>), <d-out>)
graph.edge(graph.source(<e>, compass: "e"), <e-out>)
graph.edge(graph.source(<a>), <a-out>)
},
 name: "demo",
)
#let layed-out = layout(g)
#let east = subgraph.compass(layed-out, "e")
#draw(layed-out, subgraph: east, source-style: source-style, sink-style: sink-style)

#let p = graph.build({
 graph.node(<pa>, label: [a], pos: graph.pos(y: 0, mode: "pin"))
 graph.node(<pc>, label: [c], pos: graph.pos(y: 0, mode: "pin"))
 graph.node(<pc1>, label: [c], pos: graph.pos(y: 0, mode: "pin"))
 graph.node(<pc2>, label: [c], pos: graph.pos(y: 0, mode: "pin"))
 graph.edge(graph.source(<pa>), <e0>, graph.sink(<pc>))
 graph.edge(graph.source(<pc2>), <e1>, graph.sink(<pc1>))
},
 name: "parallel demo",
)
#let parallel-edge-style(edge) = (
 offset: -0.5,
 length: 1.4,
 ratio: 0.5,
 resolve-length: "min",
 stroke: (paint: rgb("#2f6f4e"), thickness: 1.1pt, cap: "round"),
)

#let focused-base-style(edge) = (
 stroke: (paint: gray, thickness: 0.7pt, cap: "round"),
 pattern: "coil",
 pattern-amplitude: 0.14,
 pattern-wavelength: 0.45,
 pattern-coil-longitudinal-scale: 1.5,
)
#let focused-source-style(edge) = (focused-base-style(edge), parallel-edge-style(edge))
#let focused-sink-style(edge) = (focused-base-style(edge), parallel-edge-style(edge) + (
 mark: (end: ">"),
))
#draw(layout(p, g-center: 0.004, label-steps: 0,), unit: 1.25, source-style: focused-source-style, sink-style: focused-sink-style)

#let g = graph.build({
 graph.node(<a>, pos: graph.pos(x: 0, y: 0, mode: "pin"))
 graph.node(<c>, pos: graph.pos(x: 3, y: 0, mode: "pin"))
 graph.node(<d>, pos: graph.pos(x: 3, y: -1, mode: "pin"))
 graph.edge(graph.source(<a>), <ac>, graph.sink(<c>), orientation: "default")
 graph.edge(graph.source(<a>), <ad>, graph.sink(<d>), orientation: "reversed")
},
 name: "oriented marks",
)

```

```

)
#let arrow = (
 end: (symbol: ">", fill: black, anchor: "center", shorten-to: auto),
 scale: 0.75,
)
#let oriented-arrow = (
 stroke: black + 0.7pt,
 mark: arrow,
 mark-position: "center-if-dangling",
 mark-orientation: "edge",
)
#draw(layout(g), unit: 1.4, source-style: oriented-arrow, sink-style: oriented-arrow)

```

### 5.6.3.1 Parameters

```

draw(
 graph: bytes ,
 scope: dictionary ,
 unit: auto int float length ratio ,
 title: none auto content string ,
 subgraph: none bytes array ,
 debug: bool int ,
 node-radius: auto int float array ,
 node-min-radius: int float ,
 node-label-padding: int float ,
 node-fill: any ,
 node-stroke: any ,
 node-outset: auto int float ,
 node-label-style: dictionary ,
 node-style: dictionary function none ,
 node-label: auto content string function none ,
 draw-node: auto function ,
 edge-stroke: any ,
 edge-offset: int float ,
 edge-length: none int float ,
 edge-ratio: none int float ,
 edge-resolve-length: string function ,
 edge-accuracy: float ,
 edge-optimize: bool ,
 source-style: dictionary array function none ,
 sink-style: dictionary array function none ,
 edge-label: content string function none ,
 edge-label-style: dictionary function none ,
 edge-omega: float ,
 edge-trim-accuracy: float ,
 padding: none int float array dictionary ,
 debug-edge-radius: int float ,
 debug-edge-fill: any ,
 debug-edge-stroke: any ,
 debug-edge-label-fill: any ,
 subgraph-edge-style: dictionary ,
 subgraph-edge-underlay: bool
) -> content

```

#### 5.6.3.1.1 graph bytes

Graph object with positions from layout or explicit graph API pos fields.

#### 5.6.3.1.2 **scope**    dictionary

Additional values merged into node and edge callback dictionaries.

Default: `( : )`

#### 5.6.3.1.3 **unit**    `auto` or `int` or `float` or `length` or `ratio`

Coordinate length for one graph-layout unit. `auto` uses any unit stored by `graph.style`, falling back to `1em`. Numbers are interpreted as `em`.

Default: `auto`

#### 5.6.3.1.4 **title**    `none` or `auto` or `content` or `string`

Optional title displayed above the diagram. Use `auto` for the graph name.

Default: `none`

#### 5.6.3.1.5 **subgraph**    `none` or `bytes` or `array`

Optional subgraph or array of subgraphs whose half-edges are shaded. Array entries may be raw subgraphs or records like `(subgraph: sg, edge-style: (stroke: red + 2pt))`.

Default: `none`

#### 5.6.3.1.6 **debug**    `bool` or `int`

Debug level. 1 enables CeTZ canvas debug; 2 also marks edge positions.

Default: `false`

#### 5.6.3.1.7 **node-radius**    `auto` or `int` or `float` or `array`

Default CeTZ node radius. Use `auto` to fit the node label.

Default: `auto`

#### 5.6.3.1.8 **node-min-radius**    `int` or `float`

Minimum radius used when `node-radius` is `auto`.

Default: `0.16`

#### 5.6.3.1.9 **node-label-padding**    `int` or `float`

Extra canvas-unit padding around labels when `node-radius` is `auto`.

Default: `0.08`

#### 5.6.3.1.10 node-fill any

Default CeTZ node fill.

Default: white

#### 5.6.3.1.11 node-stroke any

Default CeTZ node stroke.

Default: black

#### 5.6.3.1.12 node-outset auto or int or float

Edge clearance from node centers. auto uses each node circle radius. Increase this when labels extend beyond the circle.

Default: auto

#### 5.6.3.1.13 node-label-style dictionary

Default node-label style forwarded to `cetz.draw.content`.

Default: ( : )

#### 5.6.3.1.14 node-style dictionary or function or none

Node style dictionary or callback. A callback receives node data.

Default: ( : )

#### 5.6.3.1.15 node-label auto or content or string or function or none

Node label content or callback. auto uses the node name.

Default: auto

#### 5.6.3.1.16 draw-node auto or function

CeTZ-compatible node drawing callback. auto draws the default circular node and label. A callback receives (node, box) and should return CeTZ draw elements; box contains name, center, width, height, unit, label, label-style, style, radius, and node.

Default: auto

#### 5.6.3.1.17 edge-stroke any

Default CeTZ edge stroke.

Default: `0.1em`

#### 5.6.3.1.18 edge-offset int or float

Default normal offset for edge paths. Applied to the base edge geometry before patterns; node outsets then trim the shifted path.

Default: `0`

#### 5.6.3.1.19 edge-length none or int or float

Maximum visible arc length for centered parallel edge paths. none keeps the full shifted path.

Default: `none`

#### 5.6.3.1.20 edge-ratio none or int or float

Maximum visible fraction of the base edge length for centered parallel edge paths. Combined with edge-length according to edge-resolve-length.

Default: `none`

#### 5.6.3.1.21 edge-resolve-length string or function

Resolve edge-length and edge-ratio. Accepted string values are "min"/"shorter", "max"/"longer", "length"/"fixed", "ratio"/"relative", or "none"/"full". A function receives (base-length, length, ratio).

Default: `"min"`

#### 5.6.3.1.22 edge-accuracy float

Arc-length accuracy for fitted parallel edge paths.

Default: `0.001`

#### 5.6.3.1.23 edge-optimize bool

Let Kurbo optimize the fitted parallel path.

Default: `true`

#### 5.6.3.1.24 source-style    dictionary or array or function or none

Source half-edge style dictionary, array of layer dictionaries, or callback. `mark-position`: "center-if-dangling" keeps an end marker at the paired-edge split point while centering it on dangling half edges. `mark-orientation`: "edge" makes a mark follow `edge.orientation` instead of raw path direction; reversed edges move the mark to the sink half and flip it, while undirected edges suppress it. `source-anchor` may be a CeTZ anchor name such as "north" or "south" to route this endpoint from a measured node-box anchor. By default, anchored paired edges use two smooth cubic halves through the edge layout point while preserving the endpoint anchor tangents. Set `route`: "hobby-through" to instead build a single Hobby spline through the source anchor, source guide, edge point, sink guide, and sink anchor, then split source/sink styling at the edge point. `route`: "direct" keeps the same anchored cubic routing but suppresses the default edge-position Hobby route. `route-points`: "through" also threads any layout-provided half-edge route points through that same Hobby path. `route`: "straight-through" draws the two straight force springs from source to edge position and from edge position to sink.

Default: ( : )

#### 5.6.3.1.25 sink-style    dictionary or array or function or none

Sink half-edge style dictionary, array of layer dictionaries, or callback. A callback receives edge data. `mark-position`: "center-if-dangling" has the same dangling-edge behavior as for `source-style`, and `mark-orientation`: "edge" participates in the same orientation-aware mark placement. `sink-anchor` may be a CeTZ anchor name such as "north" or "south" to route this endpoint into a measured node-box anchor. `route`: "direct" has the same meaning as in `source-style`.

Default: ( : )

#### 5.6.3.1.26 edge-label    content or string or function or none

Edge label content or callback. A callback receives edge data.

Default: none

#### 5.6.3.1.27 edge-label-style    dictionary or function or none

Edge-label style forwarded to `cetz.draw.content`; use `anchor` to choose which point of the label is placed at the layout label position. A callback receives edge data.

Default: ( : )

#### 5.6.3.1.28 edge-omega    float

Hobby curl used at the endpoints of paired edge curves.

Default: 1.0

#### 5.6.3.1.29 edge-trim-accuracy float

Optional style key for anchored source/sink routes. Set anchor-control-distance in source-style or sink-style to override the automatic guide distance used by cubic anchored routes. -> auto | int | float Arc-length accuracy for trimming edge curves at node outsets.

Default: 0.001

#### 5.6.3.1.30 padding none or int or float or array or dictionary

CeTZ canvas padding.

Default: 0.4

#### 5.6.3.1.31 debug-edge-radius int or float

Radius for edge-position markers shown at debug >= 2.

Default: 0.08

#### 5.6.3.1.32 debug-edge-fill any

Fill for edge-position markers shown at debug >= 2.

Default: rgb( "#ff9f1c" )

#### 5.6.3.1.33 debug-edge-stroke any

Stroke for edge-position markers shown at debug >= 2.

Default: rgb( "#d72638" ) + 0.35pt

#### 5.6.3.1.34 debug-edge-label-fill any

Label fill for edge-position markers shown at debug >= 2.

Default: rgb( "#7a1020" )

#### 5.6.3.1.35 subgraph-edge-style dictionary

Default CeTZ style used to shade half-edges included in subgraph. Individual subgraph records may override this with edge-style.

Default: (stroke: rgb( "#fd166" ) + 4.5pt)

#### 5.6.3.136 subgraph-edge-underlay bool

Draw subgraph shading below the normal half-edge style.

Default: `true`

## 5.7 subgraph

- `label()`
- `bits()`
- `compass()`
- `to-label()`
- `hedges()`
- `contains()`
- `complement()`
- `contains-edge()`
- `node-depths()`

### 5.7.1 label

Construct a subgraph object from a base62 label.

```
#let g = graph.build({
 graph.node(<a>)
 graph.node()
 graph.edge(graph.source(<a>, compass: "e"), graph.sink())
})
#let east = subgraph.compass(g, "e")
#let same = subgraph.label(g, subgraph.to-label(east))
#subgraph.hedges(same).len()
```

#### 5.7.1.1 Parameters

```
label(
 graph: dictionary,
 label: string
) -> bytes
```

##### 5.7.1.1.1 graph dictionary

Graph object whose half-edge set the label refers to.

##### 5.7.1.1.2 label string

Base62 subgraph label returned by `to-label` or produced by Linnest.

### 5.7.2 bits

Construct a subgraph object from a boolean hedge array.

```
#let g = graph.build({
 graph.node(<a>)
 graph.node()
 graph.edge(graph.source(<a>), graph.sink())
})
#let first = subgraph.bits(g, (true, false))
#subgraph.contains(first, 0)
```

#### 5.7.2.1 Parameters

```
bits(
 graph: dictionary,
 bits: array
) -> bytes
```

##### 5.7.2.1.1 **graph** dictionary

Graph object whose half-edge order defines the bit array.

##### 5.7.2.1.2 **bits** array

Boolean array selecting half edges by graph half-edge index.

#### 5.7.3 compass

Construct a subgraph object from a DOT compass point such as "n" or "s".

```
#let g = graph.build({
 graph.node(<a>)
 graph.node()
 graph.edge(graph.source(<a>, compass: "e"), graph.sink())
})
#subgraph.hedges(subgraph.compass(g, "e")).len()
```

#### 5.7.3.1 Parameters

```
compass(
 graph: dictionary,
 compass: string
) -> bytes
```

##### 5.7.3.1.1 **graph** dictionary

Graph object to filter by half-edge compass statement.

#### 5.7.3.1.2 compass `string`

DOT compass point such as "n", "s", "e", or "w".

### 5.7.4 to-label

Convert a subgraph object to its base62 label.

```
#let g = graph.build({
 graph.node(<a>)
 graph.node()
 graph.edge(graph.source(<a>), graph.sink())
})
#subgraph.to-label(subgraph.bits(g, (true, false)))
```

#### 5.7.4.1 Parameters

`to-label`(subgraph: `bytes`) -> `string`

##### 5.7.4.1.1 subgraph `bytes`

Subgraph object returned by this module or by `graph.cycles/graph.forests`.

### 5.7.5 hedges

Return the hedge indices included in a subgraph object.

```
#let g = graph.build({
 graph.node(<a>)
 graph.node()
 graph.edge(graph.source(<a>), graph.sink())
})
#subgraph.hedges(subgraph.bits(g, (true, false)))
```

#### 5.7.5.1 Parameters

`hedges`(subgraph: `bytes`) -> `array`

##### 5.7.5.1.1 subgraph `bytes`

Subgraph object to inspect.

### 5.7.6 contains

Test whether a subgraph object includes a hedge.

```
#let g = graph.build({
 graph.node(<a>)
 graph.node()
 graph.edge(graph.source(<a>), graph.sink())
})
#subgraph.contains(subgraph.bits(g, (true, false)), 0)
```

#### 5.7.6.1 Parameters

```
contains(
 subgraph: bytes ,
 hedge: int
) -> bool
```

##### 5.7.6.1.1 subgraph bytes

Subgraph object to inspect.

##### 5.7.6.1.2 hedge int

Half-edge index to test.

### 5.7.7 complement

Return the complement of a subgraph's selected half edges.

#### 5.7.7.1 Parameters

```
complement(
 graph: dictionary ,
 selected: bytes
) -> bytes
```

##### 5.7.7.1.1 graph dictionary

Graph object whose half-edge order defines the result.

##### 5.7.7.1.2 selected bytes

Subgraph object to invert.

### 5.7.8 contains-edge

Test whether either half-edge of an edge record is selected.

#### 5.7.8.1 Parameters

```
contains-edge(
 subgraph: bytes ,
 edge: dictionary
) -> bool
```

##### 5.7.8.1.1 subgraph bytes

Subgraph object to inspect.

##### 5.7.8.1.2 edge dictionary

Edge record from `graph.edges(...)` or draw callback data.

### 5.7.9 node-depths

Compute breadth-first node depths inside a selected edge set.

#### 5.7.9.1 Parameters

```
node-depths(
 graph: dictionary ,
 selected: bytes ,
 root: int
) -> dictionary
```

##### 5.7.9.1.1 graph dictionary

Graph object to inspect.

##### 5.7.9.1.2 selected bytes

Subgraph whose selected half edges define the traversal edges.

##### 5.7.9.1.3 root int

Root node index.

Default: 0

## 6 Rust and Python APIs

### 6.1 Rust package

The `linnet` package is organized into these principal areas:

- `half_edge` provides `HedgeGraph`, builders, the involution and edge data, subgraph types, traversal trees, and graph algorithms;
- `parser` provides DOT parsing and DOT-backed graph data;
- `half_edge::layout` provides layout helpers when the drawing feature is enabled;
- `permutation`, `tree`, and `union_find` provide supporting algorithms and stores.

The generic graph types separate edge, vertex, and half-edge payloads from the node-storage implementation. Start with the [curated Rust API](#), then follow its Rustdoc links for complete generic bounds and method signatures; choose a concrete storage type according to the workflow and mutation behavior you need.

Cargo features enable optional capabilities:

- `serde` adds serialization traits;
- `bincode` adds binary encoding support;
- `drawing` enables Linnet's layout module and its optional geometry dependency;
- `symbolica` adds Symbolica interoperability.

An item shown in an all-features API reference may not be available in a default build. Check its required feature and your `Cargo.toml` before using it.

#### Layout is not rendering

Linnet can compute layout coordinates, but it does not own a supported Rust renderer. Emit DOT for interchange, use `Clinnet` for command-line figure batches, or use `Linnest` when a Typst document owns the final drawing.

### 6.2 Standalone Python distribution

#### Distribution and import have different names

Install the Python distribution `linnet-py`, then import `linnet_py`. It requires Python 3.10 or newer. It is a standalone extension and is not a `symbolica.community` module. Its package version is independent of the Rust `linnet` version. Record both versions when diagnosing compatibility between Rust and Python code.

The Python binding focuses on DOT-backed graphs and mirrors typed identifiers and subgraphs:

```
from linnet_py import DotGraphBuilder

builder = DotGraphBuilder()
left = builder.add_node()
right = builder.add_node()
builder.add_edge(left, right)
graph = builder.build()

print(graph.dot())
```

`DotGraph` can also parse one graph from a string or file, or a set of graphs from a string. The exported classes include `Hedge`, `NodeIndex`, `EdgeIndex`, `Flow`, `Orientation`, graph attribute wrappers, `HedgePair`, `Subgraph`, `Cycle`, `OrientedCut`, `TraversalTree`, `DotGraph`, and `DotGraphBuilder`.

Mutable statement wrappers expose mapping-like access to DOT attributes. Consult the structured [Python API](#) for their lifetime and write-through behavior, and do not assume that a copied mapping remains attached to its graph.

## 7 Version history

### History coverage

Linnet and Clinnet have canonical histories. The independently versioned `linnet-py` distribution does not yet have a separate changelog in this repository.

### 7.1 Available histories

- Linnet Rust library versions
- Clinnet command-line renderer versions
- `linnet-py`: no standalone history yet

The changelog records user-visible graph, parser, drawing, and algorithm changes. In particular, users moving across releases should check index ordering, DOT serialization, subgraph behavior, and feature-gated integration changes rather than assuming a serialized graph or incidental ordering is stable.

### 7.2 Choosing the matching documentation

The latest channel may describe unreleased behavior. Documentation under `snapshots/<tag>` is fixed to a tagged repository revision. Choose a snapshot whose component information lists your Rust crate version, and check the `linnet-py` version separately when using the Python bindings.

### 7.3 Upgrade checklist

Review changes to index ordering, DOT serialization, subgraph behavior, and enabled features. Re-validate serialized graphs and any code that depends on traversal or iteration order.

### 7.4 Reproducibility record

Record the repository revision, `linnet` and `linnet-py` versions, enabled features, and the input graph format. These distinguish a library change from a parser, serializer, or binding change.

## 8 Linnet versions

This page renders the canonical Linnet changelog source as part of the Linnet website, navigation, and search.

All notable changes to this project will be documented in this file.

The format is based on [Keep a Changelog](#), and this project adheres to [Semantic Versioning](#).

### 9.1 Unreleased

#### 9.2 0.17.0 - 2026-01-28

##### 9.2.1 Other

- force-directed layout
- make the powerset iterator properly size the subset
- upgrade to all spanning *forests* generation
- Refactor PowersetIterator to support generic ID type
- double ended iterator for subsets
- remove unused crates

## 9.3 0.16.1 - 2026-01-13

### 9.3.1 Other

- Quote string values in graph and parser outputs
- revert edge, node and hedge index order to be direct (not based on sort)

## 9.4 0.16.0 - 2026-01-08

### 9.4.1 Other

- Add methods for identifying bridges and non-bridges in subgraphs
- Make trace\_unfold implementable outside of linnet

## 9.5 0.15.1 - 2026-01-05

### 9.5.1 Fixed

- fixed ordering problem in merge function and make ordering based on sorting not on direct map

### 9.5.2 Other

- update quote stripping
- Add agent context files and update project tooling
- Add graph algorithms module with topological sort and transitive ops

## 9.6 0.15.0 - 2025-12-08

### 9.6.1 Other

- Refactor Figment handling and improve data merging logic
- (*clinnet*) release v0.1.8

## 9.7 0.14.1 - 2025-12-03

### 9.7.1 Other

- Add parallel processing and directional constraints
- update test
- only swap one edge when connecting identities

## 9.8 0.14.0 - 2025-11-28

### 9.8.1 Other

- change output name and make it short, and add directional groups

## 9.9 0.13.3 - 2025-11-24

### 9.9.1 Other

- update symbolica

## **9.10 0.13.2 - 2025-11-12**

### **9.10.1 Other**

- update to symbolica 0.20
- better templates
- clinnet for nested dots to pdf
- Add Figment integration for configuration handling

## **9.11 0.13.1 - 2025-11-05**

### **9.11.1 Other**

- remove prints

## **9.12 0.12.0 - 2025-08-18**

### **9.12.1 Other**

- all tests pass
- clippy fixes
- write global data outside of graph[] scope
- Add ID=ID statements to global data when parsing
- newline after nodes
- dot serialize with name
- write graph name
- correctly parse orientations
- make iter\_edges iterate with edge\_id order not involution order

## **9.13 0.11.0 - 2025-07-29**

### **9.13.1 Other**

- serialize global data

## **9.14 0.10.0 - 2025-07-29**

### **9.14.1 Other**

- Add check graph function and extracting by nodes
- Add is\_empty to Swap trait and update implementations

## **9.15 0.9.1 - 2025-07-28**

### **9.15.1 Other**

- impl asref for newtypes

## **9.16 0.9.0 - 2025-07-27**

### **9.16.1 Other**

- Allow specifying a subgraph in the DOT format.

- Rewrite DOT output to include node, hedge and edge indices and fix bidirectional serialization
- DotGraph has concrete data. Parser now supports global edge and vertex data. Can set edge, vertex and hedge index
- add multi-graph parsing and global graph data

## 9.17 0.8.0 - 2025-07-14

### 9.17.1 Other

- clippy fix
- Rename HedgeVec to EdgeVec and introduce HedgeVec
- Add H parameter to HedgeGraph in iter\_edges and orientation
- add H to SignedCycle
- rename as\_ref to to\_ref
- Add hedge\_data to hedge\_graph
- add perm by key
- update symbolica
- Fix tree traversal and modify ForestNodeStore API to use Option

## 9.18 0.7.1 - 2025-05-24

### 9.18.1 Other

- Fix crown computation by inverting subgraph inclusion check and add inclusion for HedgePair

## 9.19 0.7.0 - 2025-05-23

### 9.19.1 Other

- unify iterator signature and remove redundant and misleading functions
- Add zenodo doi

## 9.20 0.6.3 - 2025-05-23

### 9.20.1 Other

- improve README

## 9.21 0.6.2 - 2025-05-22

### 9.21.1 Other

- remove useless check and set
- Correctly forget identification history and adds subgraph contraction

## 9.22 0.6.1 - 2025-05-22

### 9.22.1 Other

- Merge pull request #11 from [alpha100p/jules\\_wip\\_3253120278989361716](#)

## **9.23 0.6.0 - 2025-05-22**

### **9.23.1 Fixed**

- fix all tests and update symbolica
- fixes forget\_identification\_history, If you delete a subgraph you need

### **9.23.2 Other**

- Add crown functions and implement spanning tree enumeration algo based on graph contraction
- PartialEq and Eq for hedgevec
- from\_raw for hedgevec

## **9.24 0.5.2 - 2025-05-09**

### **9.24.1 Fixed**

- fix involution extraction

### **9.24.2 Other**

- Add Permutation iterators, new graph extract test

## **9.25 0.5.1 - 2025-05-05**

### **9.25.1 Other**

- Clean up debug prints and add cycle test

## **9.26 0.5.0 - 2025-05-05**

### **9.26.1 Fixed**

- fixes all around

### **9.26.2 Other**

- boundary->crown
- hairs->boundary

## **9.27 0.4.0 - 2025-04-30**

### **9.27.1 Added**

- feature flag bincode and serde

## **9.28 0.3.0 - 2025-04-30**

### **9.28.1 Fixed**

- fixed extraction for childvec forest, all tests pass and fix warnings

### **9.28.2 Other**

- adds extract to SmartHedgeVec and involution

- maps now take a FnMut
- derive more (Decode and Encode)

## 9.29 0.2.3 - 2025-04-08

### 9.29.1 Other

- unpin serde

## 9.30 0.2.2 - 2025-04-01

### 9.30.1 Fixed

- fix test snapshots

### 9.30.2 Other

- raw image

## 9.31 0.2.1 - 2025-04-01

### 9.31.1 Other

- Update README.md, logo is a github asset
- put crates badge

## 9.32 0.2.0 - 2025-04-01

### 9.32.1 Fixed

- fix cetz orientation

### 9.32.2 Other

- Create release-plz.yml

## 10 Clinnet versions

Clinnet is versioned independently from the Linnet library. Its canonical version history remains the [Clinnet changelog source](#). This rendered copy makes that history part of the manual, navigation, search, and product PDF.

All notable changes to this project will be documented in this file.

The format is based on [Keep a Changelog](#), and this project adheres to [Semantic Versioning](#).

## 11.1 Unreleased

## 11.2 0.1.8 - 2025-12-04

### 11.2.1 Other

- redo cli to have subcommands

### **11.3 0.1.7 - 2025-12-04**

#### **11.3.1 Other**

- properly forward inputs for a single figure

### **11.4 0.1.6 - 2025-12-03**

#### **11.4.1 Other**

- Add parallel processing and directional constraints
- add inputs

### **11.5 0.1.5 - 2025-11-28**

#### **11.5.1 Other**

- change output name and make it short, and add directional groups

### **11.6 0.1.4 - 2025-11-26**

#### **11.6.1 Other**

- Make figure and grid templates optional with defaults

### **11.7 0.1.3 - 2025-11-26**

#### **11.7.1 Other**

- don't overwrite existing templates
- strip quotes strings for typst

### **11.8 0.1.2 - 2025-11-24**

#### **11.8.1 Other**

- update symbolica

### **11.9 0.1.1 - 2025-11-12**

#### **11.9.1 Other**

- remove unnecessary outputs
- better templates

## API reference

The packages available in this version are listed below. Follow an item link for its complete language reference.

### linnet

Linnet's supported half-edge graph API. Algorithms are grouped in the order used by the manual and full Rustdoc.

#### Half-edge graph

Stores graph topology as paired half-edges with independent edge, vertex, and half-edge payloads.

The main graph data structure, representing a graph using the half-edge (or doubly connected edge list - DCEL) principle.

A `HedgeGraph` stores nodes and edges (represented as pairs of half-edges). It is generic over the data associated with edges (`E`), nodes/vertices (`V`), and the specific storage strategy used for nodes (`S`).

The half-edge representation allows for efficient traversal of graph topology, such as iterating around faces (in planar embeddings), finding incident edges to a node, or quickly accessing an edge's opposite half-edge.

#### # Type Parameters

- `E`: The type of data associated with each edge (or pair of half-edges).
- `V`: The type of data associated with each node (vertex).
- `S`: The node storage strategy, implementing the `NodeStorage` trait. This determines how node data and their connectivity to half-edges are stored. Defaults to `DefaultNodeStore<V>` (feature-selected; `nodestore-vec` uses `nodestore::NodeStorageVec<V>`).

```
pub struct HedgeGraph < E, V, H = NoData, S : NodeStorage < NodeData = V > =
DefaultNodeStore < V > > {}
```

**Reference feature matrix:** `nodestore-vec`, `serde`, `bincode`, `drawing`, `symbolica` (item is available without an additional gate)

#### Supported usage

```
use linnet::half_edge::HedgeGraph;
```

#### node\_store

**Kind:** Field

S

Storage for all nodes in the graph, including their data (`V`) and information about the half-edges incident to them.

The specific implementation is determined by the `S` type parameter.

Exhaustive reference · Source

#### Graph builder

Builds a valid half-edge graph while assigning typed node, edge, and hedge identifiers.

A builder for programmatically constructing [`HedgeGraph`] instances.

This builder allows for the incremental addition of nodes and edges (both paired and external/dangling) before finalizing the graph structure.

#### # Type Parameters

- `E`: The type of data to be associated with edges.
- `V`: The type of data to be associated with nodes.

#### # Example

```
```rust,ignore
use linnet::half_edge::builder::HedgeGraphBuilder;
use linnet::half_edge::involution::{Flow, Orientation};

let mut builder = HedgeGraphBuilder::<&str, &str>::new();

let node0 = builder.add_node("Node_0_Data");
let node1 = builder.add_node("Node_1_Data");
let node2 = builder.add_node("Node_2_Data");

builder.add_edge(node0, node1, "Edge_01_Data", Orientation::Default);
builder.add_external_edge(node2, "External_Edge_Data", Orientation::Undirected,
Flow::Source);

// Assuming a NodeStorage type MyNodeStore is defined and implements NodeStorageOps
// let graph: HedgeGraph<&str, &str, MyNodeStore> = builder.build();
```

pub struct HedgeGraphBuilder < E, V, H = NoData > {}
```

**Reference feature matrix:** `nodestore-vec`, `serde`, `bincode`, `drawing`, `symbolica` (item is available without an additional gate)

#### Example 1

```
use linnet::half_edge::builder::HedgeGraphBuilder;
use linnet::half_edge::involution::{Flow, Orientation};

let mut builder = HedgeGraphBuilder::<&str, &str>::new();

let node0 = builder.add_node("Node_0_Data");
let node1 = builder.add_node("Node_1_Data");
let node2 = builder.add_node("Node_2_Data");

builder.add_edge(node0, node1, "Edge_01_Data", Orientation::Default);
builder.add_external_edge(node2, "External_Edge_Data", Orientation::Undirected,
Flow::Source);

// Assuming a NodeStorage type MyNodeStore is defined and implements NodeStorageOps
// let graph: HedgeGraph<&str, &str, MyNodeStore> = builder.build();
```

#### Supported usage

```
use linnet::half_edge::builder::HedgeGraphBuilder;
```

Exhaustive reference · Source

### Bit-vector subgraph

Represents a subgraph as an efficient half-edge inclusion filter.

Represents a subgraph as an efficient half-edge inclusion filter.

```
pub type SuBitGraph = SubSet < Hedge > ;
```

**Reference feature matrix:** nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

### Supported usage

```
use linnet::half_edge::subgraph::SuBitGraph;
```

Exhaustive reference · Source

### Traversal tree

Records parent, root, and traversal information produced by breadth-first or depth-first graph walks.

Represents a traversal tree (e.g., DFS or BFS tree) derived from a [`HedgeGraph``].

This structure uses a generic [`Forest``] to store the tree topology. Each "root" in the

`Forest`` corresponds to a node in the original `HedgeGraph``, and its data is a [`TTRoot``]

enum indicating its role (Root, Child, or None) in this specific traversal. The nodes within each tree of the `Forest`` typically represent the half-edges that form the traversal path.

#### # Type Parameters

- `P``: The type of [`ForestNodeStore``] used by the underlying `Forest``. It stores `()`` as `NodeData`` because the tree nodes (representing half-edges) don't need additional data beyond their structural role in the traversal tree. Defaults to [`ParentPointerStore<()>``].

```
pub struct SimpleTraversalTree < P : ForestNodeStore < NodeData = () > =
ParentPointerStore < () > > {}
```

**Reference feature matrix:** nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

### Supported usage

```
use linnet::half_edge::tree::SimpleTraversalTree;
```

### tree\_subgraph

**Kind:** Field

SuBitGraph

A bitmask representing the set of half-edges that constitute the actual edges of this traversal tree.

Exhaustive reference · Source

### DOT-backed graph

Combines a half-edge graph with DOT attributes and parser/serializer support.

Combines a half-edge graph with DOT attributes and parser/serializer support.

```
pub struct DotGraph < N : NodeStorage < NodeData = DotVertexData > =
DefaultNodeStore < DotVertexData > > {}
```

**Reference feature matrix:** nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

### Supported usage

```
use linnet::parser::DotGraph;
```

#### global\_data

**Kind:** Field

GlobalData

Graph-level DOT metadata, including its name, payload, and attribute defaults.

#### graph

**Kind:** Field

HedgeGraph < DotEdgeData, DotVertexData, DotHedgeData, N >

Parsed half-edge topology together with its DOT edge, vertex, and half-edge data.

Exhaustive reference · Source

#### Half-edge pairing

Represents an unpaired external half-edge, a paired full edge, or an edge split across a subgraph boundary.

Describes the pairing status of a half-edge, indicating whether it forms a complete edge with another half-edge, is a dangling/external edge, or is part of a "split" edge in a subgraph context.

```
pub enum HedgePair {}
```

**Reference feature matrix:** nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

### Supported usage

```
use linnet::half_edge::involution::HedgePair;
```

#### Unpaired

**Kind:** Variant

Represents a half-edge that is unpaired (dangling or external).

#### hedge

**Kind:** Field

Hedge

The half-edge itself.

#### flow

**Kind:** Field

Flow

The [``Flow``] (directionality) of this unpaired half-edge.

#### Paired

**Kind:** Variant

Represents two half-edges that are paired together to form a complete, internal, undirected edge.

#### **source**

**Kind:** Field

Hedge

The half-edge considered the "source" in this pair.

#### **sink**

**Kind:** Field

Hedge

The half-edge considered the "sink" in this pair, opposite to ``source``.

#### **Split**

**Kind:** Variant

Represents an edge that is "split" in the context of a subgraph. This typically means one half-edge is inside the subgraph and its opposite is outside, or vice-versa.

#### **source**

**Kind:** Field

Hedge

The half-edge on one side of the split (often the one inside a subgraph).

#### **sink**

**Kind:** Field

Hedge

The half-edge on the other side of the split (often the one outside a subgraph).

#### **split**

**Kind:** Field

Flow

Indicates which side of the original conceptual edge (``source`` or ``sink`` of the pair) is considered "split off" or external. For example, if ``split` == Flow::Source`, then the ``source`` half-edge is effectively dangling from the perspective of the other side of the split.

Exhaustive reference · Source

#### **Intrinsic half-edge flow**

Marks a half-edge as the intrinsic source or sink within its full edge.

Represents the underlying, intrinsic directionality of a half-edge relative to the full edge it is part of.

Each full edge is composed of two half-edges. One is designated as the ``Source`` and the other as the ``Sink``. This is a fundamental property used by the ``Involution`` to pair half-edges correctly.

```
pub enum Flow {}
```

**Reference feature matrix:** nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

### Supported usage

```
use linnet::half_edge::involution::Flow;
```

#### Source

**Kind:** Variant

Indicates this half-edge is the "source" part of its full edge. This is often visualized as the outgoing part of a directed edge if the edge's `Orientation` is `Default`.

#### Sink

**Kind:** Variant

Indicates this half-edge is the "sink" part of its full edge. This is often visualized as the incoming part of a directed edge if the edge's `Orientation` is `Default`.

Exhaustive reference · Source

### Edge orientation

Records default, reversed, or undirected presentation independently of an edge's intrinsic half-edge flow.

Represents the superficial or conventional orientation of an edge.

This orientation can be distinct from the underlying `Flow` of the half-edges that constitute the edge. It's often used for display or for algorithms where a conventional direction is needed.

```
pub enum Orientation {}
```

**Reference feature matrix:** nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

### Supported usage

```
use linnet::half_edge::involution::Orientation;
```

#### Default

**Kind:** Variant

The edge's orientation aligns with the underlying flow of its "source" half-edge (e.g., if half-edge A is `Flow::Source` and points to B, a `Default` orientation means the edge is A -> B).

#### Reversed

**Kind:** Variant

The edge's orientation is opposite to the underlying flow of its "source" half-edge (e.g., if half-edge A is `Flow::Source` and points to B, a `Reversed` orientation means the edge is B -> A).

#### Undirected

**Kind:** Variant

The edge is considered undirected, regardless of the underlying half-edge flows.

Exhaustive reference · Source

### Half-edge subset interface

Defines the core inclusion queries, iteration, joining, and labeling contract shared by half-edge subset representations.

The core trait defining the properties and behaviors of a subgraph representation.

A subgraph is fundamentally a selection of half-edges from a larger [`HedgeGraph`]. This trait provides methods for querying the subgraph's contents, its relationship to the parent graph, and performing common operations.

Implementors must also satisfy ``Clone``, ``Eq``, ``Hash``, and various [``Inclusion``] bounds.

```
pub trait SubSetLike < ID = Hedge > : Clone + Eq + Hash + Inclusion < Self > +
 Inclusion < Self :: Base > + Inclusion < std :: ops :: Range < ID > > +
 Inclusion < std :: ops :: RangeToInclusive < ID > > + Inclusion < std :: ops
 :: RangeInclusive < ID > > + Inclusion < std :: ops :: RangeTo < ID > > +
 Inclusion < std :: ops :: RangeFrom < ID > > + Inclusion < ID > {}
```

**Reference feature matrix:** `nodestore-vec`, `serde`, `bincode`, `drawing`, `symbolica` (item is available without an additional gate)

### Supported usage

```
use linnet::half_edge::subgraph::SubSetLike;
```

#### Base

**Kind:** `AssociatedType`

```
type Base : SubSetLike < ID > ;
```

Canonical subset representation returned by [``SubSetLike::included``].

#### BaseIter

**Kind:** `AssociatedType`

```
type BaseIter < 'a > : Iterator < Item = ID > + DoubleEndedIterator where Self
: 'a, ID : 'a;
```

Double-ended iterator over the included IDs.

#### has\_greater

**Kind:** `Method`

```
fn has_greater(& self, hedge : ID) -> bool { ... }
```

Contains an id with index  $\geq$  hedge

#### has\_lesser

**Kind:** `Method`

```
fn has_lesser(& self, hedge : ID) -> bool { ... }
```

Contains an id with index  $<$  hedge

#### join

**Kind:** `Method`

```
fn join(mut self, other : Self) -> Self { ... }
```

joins two subsets into one, (this is not union)

### **join\_mut**

**Kind:** Method

```
fn join_mut(& mut self, other : Self);
```

Appends all included half-edges at the end of other

### **string\_label**

**Kind:** Method

```
fn string_label(& self) -> String;
```

Returns a compact textual label representing this selection.

### **symbol**

**Kind:** Method

```
fn symbol(& self) -> symbolica :: atom :: Symbol { ... }
```

Returns a Symbolica symbol whose name and formatter encode this selection's label.

### **from\_base62**

**Kind:** AssociatedFunction

```
fn from_base62(label : & str, size : usize) -> Option < Self >;
```

Attempts to decode an implementation-specific compact label for a universe of `size` IDs.

Returns `None` when decoding is unsupported or the label cannot represent that universe.

### **included\_iter**

**Kind:** Method

```
fn included_iter(& self) -> Self :: BaseIter < '_ >;
```

Iterates over every ID included in the selection.

### **included**

**Kind:** Method

```
fn included(& self) -> & Self :: Base;
```

Returns a simple Self::Base of all included hedges

### **size**

**Kind:** Method

```
fn size(& self) -> usize;
```

Number of half-edges in the graph this is a subgraph of

### **n\_included**

**Kind:** Method

```
fn n_included(& self) -> usize;
```

Number of half-edges included in the subgraph

### **empty**

**Kind:** AssociatedFunction

```
fn empty(size : usize) -> Self;
```

Creates a selection containing no IDs in a universe of `size` IDs.

### **is\_empty**

**Kind:** Method

```
fn is_empty(& self) -> bool;
```

Returns whether the selection contains no IDs.

Exhaustive reference · Source

### **Half-edge subset operations**

Provides in-place and value-returning intersection, union, symmetric-difference, and subtraction operations.

Defines binary set operations between a subgraph and another subset-like value.

# Type Parameters

- `Other`: The type of the other subgraph to perform operations with. Defaults to `Self`.

```
pub trait SubSetOps < ID = Hedge, Other : SubSetLike < ID > = Self > :
 SubSetLike < ID > {}
```

**Reference feature matrix:** nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

### **Supported usage**

```
use linnet::half_edge::subgraph::SubSetOps;
```

### **intersect\_with**

**Kind:** Method

```
fn intersect_with(& mut self, other : & Other);
```

Retains only IDs included by both subsets.

### **union\_with**

**Kind:** Method

```
fn union_with(& mut self, other : & Other);
```

Adds every ID included by `other`.

### **sym\_diff\_with**

**Kind:** Method

```
fn sym_diff_with(& mut self, other : & Other);
```

Retains IDs included by exactly one of the two subsets.

### **empty\_intersection**

**Kind:** Method

```
fn empty_intersection(& self, other : & Other) -> bool;
```

Returns whether the subsets have no included ID in common.

### **empty\_union**

**Kind:** Method

```
fn empty_union(& self, other : & Other) -> bool;
```

Returns whether neither subset includes any ID.

### **intersection**

**Kind:** Method

```
fn intersection(& self, other : & Other) -> Self { ... }
```

Returns the IDs included by both subsets without modifying either input.

### **union**

**Kind:** Method

```
fn union(& self, other : & Other) -> Self { ... }
```

Returns the IDs included by either subset without modifying either input.

### **sym\_diff**

**Kind:** Method

```
fn sym_diff(& self, other : & Other) -> Self { ... }
```

Returns the IDs included by exactly one subset without modifying either input.

### **subtract\_with**

**Kind:** Method

```
fn subtract_with(& mut self, other : & Other);
```

Removes every ID included by `other`.

### **subtract**

**Kind:** Method

```
fn subtract(& self, other : & Other) -> Self { ... }
```

Returns the IDs in `self` but not `other` without modifying either input.

Exhaustive reference · Source

### **Subset inclusion queries**

Abstracts full containment and non-empty intersection checks between a subgraph and another value.

A trait for checking inclusion or intersection relationships between a subgraph and another entity `T`.

This is used to determine if a subgraph contains certain elements (like specific hedges, ranges of hedges, or other subgraphs) or overlaps with them.

# Type Parameters

- `T`: The type of the entity to check for inclusion or intersection with.

```
pub trait Inclusion < T > {}
```

**Reference feature matrix:** nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

### **Supported usage**

```
use linnet::half_edge::subgraph::Inclusion;
```

### **includes**

**Kind:** Method

```
fn includes(& self, other : & T) -> bool;
```

Checks if this subgraph fully includes/contains the `other` entity.

### **intersects**

**Kind:** Method

```
fn intersects(& self, other : & T) -> bool;
```

Checks if this subgraph has any common elements with the `other` entity.

[Exhaustive reference](#) · [Source](#)

### **Mutable base subgraph**

Combines graph-aware subset behavior and mutation with construction from edge filters or half-edge iterators.

Combines graph-aware subset behavior and mutation with construction from edge filters or half-edge iterators.

```
pub trait BaseSubgraph : SubGraphLike + ModifySubSet < HedgePair > +
ModifySubSet < Hedge > + SubSetOps < Hedge > {}
```

**Reference feature matrix:** `nodestore-vec`, `serde`, `bincode`, `drawing`, `symbolica` (item is available without an additional gate)

### **Supported usage**

```
use linnet::half_edge::subgraph::BaseSubgraph;
```

### **from\_filter**

**Kind:** AssociatedFunction

```
fn from_filter < E, V, H, N : NodeStorageOps < NodeData = V > , F : FnMut(& E)
-> bool > (graph : & HedgeGraph < E, V, H, N > , filter : F,) -> Self;
```

Selects all half-edges belonging to edges whose data satisfies `filter`.

### **from\_hedge\_iter**

**Kind:** AssociatedFunction

```
fn from_hedge_iter < I : Iterator < Item = Hedge > > (iter : I, len : usize)
-> Self;
```

Creates a base subgraph containing all hedges yielded by the iterator.  
`len` specifies the total number of possible hedges in the graph context.

[Exhaustive reference](#) · [Source](#)

### **Graph-aware subgraph interface**

Extends half-edge subsets with graph coverage, edge counts, external hairs, and DOT rendering.

Extends half-edge subsets with graph coverage, edge counts, external hairs, and DOT rendering.

```
pub trait SubGraphLike : SubSetLike < Hedge > + Inclusion < HedgePair > {}
```

**Reference feature matrix:** `nodestore-vec`, `serde`, `bincode`, `drawing`, `symbolica` (item is available without an additional gate)

### **Supported usage**

```
use linnet::half_edge::subgraph::SubGraphLike;
```

## **covers**

**Kind:** Method

```
fn covers < E, V, H, N : NodeStorageOps < NodeData = V > , S : ModifySubSet < Hedge > + SubSetOps < Hedge > + SubSetLike < Hedge > , >
(& self, graph : & HedgeGraph < E, V, H, N > ,) -> S { ... }
```

maximal graph that contains all nodes of the subgraph

## **background\_color**

**Kind:** Method

```
fn background_color(& self, hedge_pair : Option < HedgePair >) -> Option < String > { ... }
```

Chooses the DOT background color used to distinguish the subgraph boundary.

The default colors split and absent edge pairs gray and leaves complete pairs uncolored.

## **nedges**

**Kind:** Method

```
fn nedges < E, V, H, N : NodeStorageOps < NodeData = V > >
(& self, graph : & HedgeGraph < E, V, H, N > ,) -> usize;
```

Number of full edges included in the subgraph

## **dot\_fmt**

**Kind:** Method

```
fn dot_fmt < W : std :: fmt :: Write, E, V, H, N : NodeStorageOps < NodeData = V > , Str : AsRef < str > >
(& self, writer : & mut W, graph : & HedgeGraph < E, V, H, N > , graph_info : Str, hedge_attr : & impl Fn(& H) -> Option < String > , edge_attr : & impl Fn(& E) -> Option < String > , node_attr : & impl Fn(& V) -> Option < String > ,) -> Result < (), std :: fmt :: Error > { ... }
```

Writes a DOT rendering of this subgraph to a formatting sink.

The attribute callbacks supply optional DOT attributes for half-edges, edges, and nodes.

## **dot\_io**

**Kind:** Method

```
fn dot_io < W : std :: io :: Write, E, V, H, N : NodeStorageOps < NodeData = V > , Str : AsRef < str > >
(& self, writer : & mut W, graph : & HedgeGraph < E, V, H, N > , graph_info : Str, hedge_attr : & impl Fn(& H) -> Option < String > , edge_attr : & impl Fn(& E) -> Option < String > , node_attr : & impl Fn(& V) -> Option < String > ,) -> Result < (), std :: io :: Error > { ... }
```

Writes a DOT rendering of this subgraph to an I/O sink.

The attribute callbacks supply optional DOT attributes for half-edges, edges, and nodes.

## **hairs**

**Kind:** Method

```
fn hairs(& self, node : impl Iterator < Item = Hedge >) -> SuBitGraph { ... }
```

Selects the half-edges that occur in both this subgraph and `node`.

[Exhaustive reference](#) · [Source](#)

### Node-storage interface

Defines a graph's node data, transformed storage family, neighbor representation, and neighbor iterator.

The fundamental trait defining the interface for a node storage backend in a [`HedgeGraph`].

This trait abstracts the way nodes and their incident half-edges are stored.

Different

implementations can offer various performance trade-offs for accessing and manipulating node data and topology.

```
pub trait NodeStorage : Sized {}
```

**Reference feature matrix:** `nodestore-vec`, `serde`, `bincode`, `drawing`, `symbolica` (item is available without an additional gate)

### Supported usage

```
use linnet::half_edge::nodestore::NodeStorage;
```

#### NodeData

**Kind:** `AssociatedType`

```
type NodeData;
```

The type of data stored for each node in the graph.

#### Storage

**Kind:** `AssociatedType`

```
type Storage < N > ;
```

A generic way to refer to this storage type when it holds node data of type `N`.

This is often used in `NodeStorageOps` for methods that transform node data.

#### Neighbors

**Kind:** `AssociatedType`

```
type Neighbors : SubSetLike;
```

The type used to represent the collection of half-edges incident to a single node.

This must implement the [`SubSetLike`] trait.

#### NeighborsIter

**Kind:** `AssociatedType`

```
type NeighborsIter < 'a > : ExactSizeIterator < Item = Hedge > + Clone where
Self : 'a;
```

An iterator that yields the [`Hedge`] identifiers incident to a node.

It must be cloneable and provide an exact size hint.

[Exhaustive reference](#) · [Source](#)

### Mutable node-storage operations

Adds construction, extraction, node identification, validation, traversal, and data mapping to a node-storage backend.

Defines a comprehensive set of operations for a node storage backend within a [`HedgeGraph`].

This trait extends [`NodeStorage`] by adding methods for modifying the storage (e.g., extending, extracting, deleting, identifying nodes), building it, iterating over nodes, and mapping node data. It abstracts over the specific data structures used to store node information and their incident half-edges.

```
pub trait NodeStorageOps : NodeStorage + Swap < Hedge > + Swap < NodeIndex >
{}
```

**Reference feature matrix:** `nodestore-vec`, `serde`, `bincode`, `drawing`, `symbolica` (item is available without an additional gate)

### Supported usage

```
use linnet::half_edge::nodestore::NodeStorageOps;
```

### OpStorage

**Kind:** `AssociatedType`

```
type OpStorage < N > : NodeStorageOps < NodeData = N > ;
```

The type of storage returned after mapping operations that might change the node data type.

### Base

**Kind:** `AssociatedType`

```
type Base : BaseSubgraph;
```

The base type of subgraph used by this node storage (e.g., a `BitVec` to represent a set of hedges).

### extend

**Kind:** `Method`

```
fn extend(self, other : Self) -> Self;
```

Combines two stores using the implementation's index-space semantics.

Callers must not assume embedded index-valued data is remapped unless the concrete storage implementation documents that behavior.

### new\_nodevec

**Kind:** `Method`

```
fn new_nodevec < 'a, V2 >
(& 'a self, node_map : impl
FnMut(NodeIndex, Self :: NeighborsIter < 'a > , & 'a Self :: NodeData) -> V2,)
-> NodeVec < V2 >;
```

Maps each node's ID, neighbors, and borrowed data into a dense [`NodeVec`].

### extend\_mut

**Kind:** `Method`

```
fn extend_mut(& mut self, other : Self);
```

Appends a store using the implementation's index-space semantics.

Callers must not assume embedded index-valued data is remapped unless the concrete storage implementation documents that behavior.

### **extract**

**Kind:** Method

```
fn extract < S : SubSetLike < Base = Self :: Base > , V2 >
(& mut self, subgraph : & S, split_node : impl FnMut(& Self :: NodeData) ->
V2, owned_node : impl FnMut(Self :: NodeData) -> V2,) -> Self :: OpStorage <
V2 >;
```

Moves the selected half-edges into a new store and removes them from `self`.

`split\_node` maps data for nodes crossed by the selection boundary, while  
`owned\_node` maps data for nodes wholly inside the selection. Implementations may  
compact both stores and thereby change hedge and node indices.

### **extract\_nodes**

**Kind:** Method

```
fn extract_nodes(& mut self, nodes : impl IntoIterator < Item = NodeIndex >)
-> (Self :: Base, Self);
```

Moves complete nodes and their incident half-edges into a separate store.

The returned subset records the selected half-edges.

### **delete**

**Kind:** Method

```
fn delete < S : SubSetLike < Base = Self :: Base > >
(& mut self, subgraph : & S);
```

Removes the selected half-edges and compacts the remaining node storage.

### **identify\_nodes**

**Kind:** Method

```
fn
identify_nodes(& mut self, nodes : & [NodeIndex], node_data_merge : Self ::
NodeData) -> NodeIndex;
```

Identifies nodes, essentially turning them into a single node  
Invalidates all previous NodeIndex values if using VecNodeStore.  
Does not invalidate if using Forest as NodeStore

### **forget\_identification\_history**

**Kind:** Method

```
fn forget_identification_history(& mut self) -> NodeVec < Self :: NodeData >;
```

Drops inactive nodes retained by identification operations and returns their data.

### **to\_forest**

**Kind:** Method

```
fn to_forest < U, H > (& self, map_data : impl Fn(& Self :: NodeData) -> U,)
-> Forest < U, ParentPointerStore < H > >;
```

Copies the topology into a parent-pointer forest while mapping borrowed node data.

## **build**

**Kind:** AssociatedFunction

```
fn build < I : IntoIterator < Item = HedgeNodeBuilder < Self :: NodeData > > >
(nodes : I, n_hedges : usize,) -> Self;
```

Builds a store from node descriptions that partition `n\_hedges` half-edges.

## **build\_with\_mapping**

**Kind:** AssociatedFunction

```
fn build_with_mapping < I : IntoIterator < Item = HedgeNodeBuilder < ND > > ,
ND >
(nodes : I, n_hedges : usize, map_data : impl FnMut(ND) -> Self :: NodeData,)
-> Self;
```

Builds a store from a half-edge partition while converting each node's data.

## **add\_dangling\_edge**

**Kind:** Method

```
fn add_dangling_edge(self, source : NodeIndex) -> Result < Self,
HedgeGraphError >;
```

Appends an unpaired half-edge incident to `source`.

## **random**

**Kind:** AssociatedFunction

```
fn random(sources : & [Self :: Neighbors], sinks : & [Self :: Neighbors]) ->
Self where Self :: NodeData : Default;
```

Builds nodes from source and sink neighbor sets, assigning default data to each node.

## **drain**

**Kind:** Method

```
fn drain(self) -> impl Iterator < Item = (NodeIndex, Self :: NodeData) >;
```

Consumes the store and yields each current node ID with its owned data.

## **iter**

**Kind:** Method

```
fn iter(& self) -> impl Iterator < Item = (NodeIndex, & Self :: NodeData) >;
```

Iterates over current node IDs and borrowed node data.

## **check\_and\_set\_nodes**

**Kind:** Method

```
fn check_and_set_nodes(& mut self) -> Result < (), HedgeGraphError >;
```

Ensures backend-maintained node mappings are ready and consistent.

## **check\_nodes**

**Kind:** Method

```
fn check_nodes(& self) -> Result < (), HedgeGraphError >;
```

Checks that every half-edge belongs to exactly one consistent node.

## **map\_data\_ref\_graph**

**Kind:** Method

```
fn map_data_ref_graph < 'a, E, V2, H >
(& 'a self, graph : & 'a HedgeGraph < E, Self :: NodeData, H, Self > ,
node_map : impl
FnMut(& 'a HedgeGraph < E, Self :: NodeData, H, Self > , Self :: NeighborsIter
< 'a > , & 'a Self :: NodeData,) -> V2,) -> Self :: OpStorage < V2 >;
```

Maps borrowed node data while retaining topology and exposing the complete graph.

#### **map\_data\_ref\_mut\_graph**

**Kind:** Method

```
fn map_data_ref_mut_graph < 'a, V2 >
(& 'a mut self, node_map : impl
FnMut(Self :: NeighborsIter < 'a > , & 'a mut Self :: NodeData) -> V2,) ->
Self :: OpStorage < V2 >;
```

Maps node data through mutable references while retaining topology.

#### **map\_data\_ref\_graph\_result**

**Kind:** Method

```
fn map_data_ref_graph_result < 'a, E, V2, H, Er >
(& 'a self, graph : & 'a HedgeGraph < E, Self :: NodeData, H, Self > ,
node_map : impl
FnMut(& 'a HedgeGraph < E, Self :: NodeData, H, Self > , Self :: NeighborsIter
< 'a > , & 'a Self :: NodeData,) -> Result < V2, Er > ,) -> Result < Self ::
OpStorage < V2 > , Er >;
```

Fallible counterpart of [``NodeStorageOps::map_data_ref_graph``].

#### **map\_data\_graph**

**Kind:** Method

```
fn map_data_graph < 'a, V2 >
(self, involution : & 'a Involution < EdgeIndex > , f : impl
FnMut(& 'a Involution < EdgeIndex > , NodeIndex, Self :: NodeData) -> V2,) ->
Self :: OpStorage < V2 >;
```

Consumes the store and maps owned node data with access to the edge involution and node ID.

#### **map\_data\_graph\_result**

**Kind:** Method

```
fn map_data_graph_result < 'a, V2, Err >
(self, involution : & 'a Involution < EdgeIndex > , f : impl
FnMut(& 'a Involution < EdgeIndex > , NodeIndex, Self :: NodeData) -> Result <
V2, Err > ,) -> Result < Self :: OpStorage < V2 > , Err >;
```

Fallible counterpart of [``NodeStorageOps::map_data_graph``].

#### **iter\_node\_id**

**Kind:** Method

```
fn iter_node_id(& self) -> impl Iterator < Item = NodeIndex > { ... }
```

Iterates over the store's dense node-index range.

#### **iter\_nodes**

**Kind:** Method

```
fn iter_nodes(& self,) -> impl Iterator < Item =
(NodeIndex, Self :: NeighborsIter < '_ > , & Self :: NodeData) >;
Iterates over each node's ID, neighbors, and borrowed data.
```

#### **iter\_nodes\_mut**

**Kind:** Method

```
fn iter_nodes_mut(& mut self,) -> impl Iterator < Item =
(NodeIndex, Self :: NeighborsIter < '_ > , & mut Self :: NodeData) >;
Iterates over each node's ID, neighbors, and mutably borrowed data.
```

#### **node\_id\_ref**

**Kind:** Method

```
fn node_id_ref(& self, hedge : Hedge) -> NodeIndex;
Retrieves the [NodeIndex] that the given [Hedge] is incident to (or originates
from).
```

#### **get\_neighbor\_iterator**

**Kind:** Method

```
fn get_neighbor_iterator(& self, node_id : NodeIndex) -> Self :: NeighborsIter
< '_ >;
Returns an iterator over the half-edges incident to the specified node_id.
```

#### **get\_node\_data**

**Kind:** Method

```
fn get_node_data(& self, node_id : NodeIndex) -> & Self :: NodeData;
Gets a reference to the data associated with node_id.
```

#### **get\_node\_data\_mut**

**Kind:** Method

```
fn get_node_data_mut(& mut self, node_id : NodeIndex) -> & mut Self ::
NodeData;
Gets a mutable reference to the data associated with node_id.
```

Exhaustive reference · Source

### **linnet-py**

Exports registered by linnet-py.

#### **Cycle**

Cycle represented as a subgraph and optional loop count.

Cycle represented as a subgraph and optional loop count.

```
class Cycle:
```

**Requires features:** extension-module, abi3-py310

#### **filter**

**Kind:** Getter

```
def filter(self) -> Subgraph:
Subgraph filter for this cycle.
```

**loop\_count****Kind:** Getter

```
def loop_count(self) -> Optional[int]:
```

Optional loop count for this cycle.

**\_\_repr\_\_****Kind:** Method

```
def __repr__(self) -> str:
```

Debug-style representation.

Exhaustive reference · Source

**DotEdgeData**

DOT edge data (attributes + optional edge id).

DOT edge data (attributes + optional edge id).

```
class DotEdgeData:
```

**Requires features:** extension-module, abi3-py310

**statements****Kind:** Getter

```
def statements(self) -> EdgeStatements:
```

Edge attributes as a dict-like proxy.

**local\_statements****Kind:** Getter

```
def local_statements(self) -> EdgeStatements:
```

Local edge attributes as a dict-like proxy.

**edge\_id****Kind:** Getter

```
def edge_id(self) -> Optional[EdgeIndex]:
```

Optional edge id.

**\_\_new\_\_****Kind:** AssociatedFunction

```
def __new__(cls, statements: Optional[Mapping[str, str]] = None, local_statements: Optional[Mapping[str, str]] = None, edge_id: Optional[int] = None) -> DotEdgeData:
```

Create edge data from fields.

**statements****Kind:** Parameter

Optional[Mapping[str, str]]

**Default:** None

**local\_statements****Kind:** Parameter

Optional[Mapping[str, str]]

**Default:** None

**edge\_id**

**Kind:** Parameter

Optional[int]

**Default:** None

**add\_statement**

**Kind:** Method

```
def add_statement(self, key: str, value: str) -> None:
```

Insert or update an attribute statement.

**key**

**Kind:** Parameter

str

**value**

**Kind:** Parameter

str

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> str:
```

Debug-style representation.

Exhaustive reference · Source

**DotGraph**

DOT-backed hedge graph.

DOT-backed hedge graph.

```
class DotGraph:
```

**Requires features:** extension-module, abi3-py310

**global\_data**

**Kind:** Getter

```
def global_data(self) -> GlobalData:
```

Global graph attributes.

**global\_data**

**Kind:** Setter

```
def global_data(self, value: GlobalData) -> None:
```

**value**

**Kind:** Parameter

GlobalData

**from\_string**

**Kind:** AssociatedFunction

```
def from_string(cls, s: str) -> DotGraph:
```

Parse a DOT string into a graph.

**s**

**Kind:** Parameter

str

**from\_string\_set**

**Kind:** AssociatedFunction

```
def from_string_set(cls, s: str) -> list[DotGraph]:
```

Parse a DOT string into multiple graphs.

**s**

**Kind:** Parameter

str

**from\_file**

**Kind:** AssociatedFunction

```
def from_file(cls, path: str) -> DotGraph:
```

Parse a DOT file into a graph.

**path**

**Kind:** Parameter

str

**debug\_dot**

**Kind:** Method

```
def debug_dot(self) -> str:
```

Serialize graph to DOT for debugging.

**dot**

**Kind:** Method

```
def dot(self) -> str:
```

Serialize the full graph to DOT.

**dot\_of**

**Kind:** Method

```
def dot_of(self, subgraph: Subgraph) -> str:
```

Serialize a subgraph to DOT.

**subgraph**

**Kind:** Parameter

Subgraph

**n\_nodes**

**Kind:** Method

```
def n_nodes(self) -> int:
```

Number of nodes.

#### **n\_edges**

**Kind:** Method

```
def n_edges(self) -> int:
```

Number of edges.

#### **n\_hedges**

**Kind:** Method

```
def n_hedges(self) -> int:
```

Number of hedges.

#### **n externals**

**Kind:** Method

```
def n_externals(self) -> int:
```

Number of external hedges.

#### **n\_internals**

**Kind:** Method

```
def n_internals(self) -> int:
```

Number of internal hedges.

#### **full\_filter**

**Kind:** Method

```
def full_filter(self) -> Subgraph:
```

Subgraph including all hedges.

#### **empty\_subgraph**

**Kind:** Method

```
def empty_subgraph(self) -> Subgraph:
```

Empty subgraph of this graph.

#### **iter\_edges**

**Kind:** Method

```
def iter_edges(self) -> list[tuple[HedgePair, EdgeIndex, EdgeData]]:
```

Iterate edges in the full graph.

#### **iter\_edges\_of**

**Kind:** Method

```
def iter_edges_of(self, subgraph: Subgraph) -> list[tuple[HedgePair, EdgeIndex, EdgeData]]:
```

Iterate edges within a subgraph.

#### **subgraph**

**Kind:** Parameter

Subgraph

**iter\_nodes****Kind:** Method

```
def iter_nodes(self) -> list[tuple[NodeIndex, list[Hedge], DotVertexData]]:
```

Iterate nodes in the full graph.

**iter\_nodes\_of****Kind:** Method

```
def iter_nodes_of(self, subgraph: Subgraph) -> list[tuple[NodeIndex, list[Hedge], DotVertexData]]:
```

Iterate nodes within a subgraph.

**subgraph****Kind:** Parameter

Subgraph

**connected\_components****Kind:** Method

```
def connected_components(self, subgraph: Subgraph) -> list[Subgraph]:
```

Connected components of a subgraph.

**subgraph****Kind:** Parameter

Subgraph

**count\_connected\_components****Kind:** Method

```
def count_connected_components(self, subgraph: Subgraph) -> int:
```

Count connected components.

**subgraph****Kind:** Parameter

Subgraph

**is\_connected****Kind:** Method

```
def is_connected(self, subgraph: Subgraph) -> bool:
```

Whether a subgraph is connected.

**subgraph****Kind:** Parameter

Subgraph

**depth\_first\_traverse****Kind:** Method

```
def depth_first_traverse(self, subgraph: Subgraph, root_node: NodeIndex, include_hedge: Optional[Hedge]) -> TraversalTree:
```

Depth-first traversal from a root node.

**subgraph****Kind:** Parameter

Subgraph

**root\_node****Kind:** Parameter

NodeIndex

**include\_hedge****Kind:** Parameter

Optional[Hedge]

**breadth\_first\_traverse****Kind:** Method

```
def breadth_first_traverse(self, subgraph: Subgraph, root_node: NodeIndex,
include_hedge: Optional[Hedge]) -> TraversalTree:
```

Breadth-first traversal from a root node.

**subgraph****Kind:** Parameter

Subgraph

**root\_node****Kind:** Parameter

NodeIndex

**include\_hedge****Kind:** Parameter

Optional[Hedge]

**bridges****Kind:** Method

```
def bridges(self) -> Subgraph:
```

Bridges in the full graph.

**bridges\_of****Kind:** Method

```
def bridges_of(self, subgraph: Subgraph) -> Subgraph:
```

Bridges within a subgraph.

**subgraph****Kind:** Parameter

Subgraph

**cycle\_basis****Kind:** Method

```
def cycle_basis(self) -> tuple[list[Cycle], Subgraph]:
```

Cycle basis of the full graph.

#### **cycle\_basis\_of**

**Kind:** Method

```
def cycle_basis_of(self, subgraph: Subgraph) -> tuple[list[Cycle], Subgraph]:
```

Cycle basis of a subgraph.

#### **subgraph**

**Kind:** Parameter

Subgraph

#### **all\_spanning\_forests**

**Kind:** Method

```
def all_spanning_forests(self) -> list[Subgraph]:
```

All spanning forests of the full graph.

#### **all\_spanning\_forests\_of**

**Kind:** Method

```
def all_spanning_forests_of(self, subgraph: Subgraph) -> list[Subgraph]:
```

All spanning forests of a subgraph.

#### **subgraph**

**Kind:** Parameter

Subgraph

#### **combine\_to\_single\_hedgenode**

**Kind:** Method

```
def combine_to_single_hedgenode(self, nodes: Sequence[NodeIndex]) -> HedgeNode:
```

Combine nodes into a single hedge node.

#### **nodes**

**Kind:** Parameter

Sequence[NodeIndex]

#### **all\_cuts**

**Kind:** Method

```
def all_cuts(self, source: HedgeNode, target: HedgeNode) -> list[tuple[Subgraph,
OrientedCut, Subgraph]]:
```

All cuts between two hedge nodes.

#### **source**

**Kind:** Parameter

HedgeNode

#### **target**

**Kind:** Parameter

HedgeNode

**all\_cuts\_from\_ids****Kind:** Method

```
def all_cuts_from_ids(self, source: Sequence[NodeIndex], target: Sequence[NodeIndex])
-> list[tuple[Subgraph, OrientedCut, Subgraph]]:
```

All cuts between two sets of node indices.

**source****Kind:** Parameter

Sequence[NodeIndex]

**target****Kind:** Parameter

Sequence[NodeIndex]

**contract\_subgraph****Kind:** Method

```
def contract_subgraph(self, subgraph: Subgraph, node_data_merge:
Optional[DotVertexData] = None) -> None:
```

Contract a subgraph into a single node, deleting its edges.

**subgraph****Kind:** Parameter

Subgraph

**node\_data\_merge****Kind:** Parameter

Optional[DotVertexData]

**Default:** None**join****Kind:** Method

```
def join(self, other: DotGraph, matching_fn: Any, merge_fn: Any) -> DotGraph:
```

Join two graphs, matching dangling edges via a Python callback.

**other****Kind:** Parameter

DotGraph

**matching\_fn****Kind:** Parameter

Any

**merge\_fn****Kind:** Parameter

Any

**join\_mut****Kind:** Method

```
def join_mut(self, other: DotGraph, matching_fn: Any, merge_fn: Any) -> None:
```

In-place join, matching dangling edges via a Python callback.

**other**

**Kind:** Parameter

DotGraph

**matching\_fn**

**Kind:** Parameter

Any

**merge\_fn**

**Kind:** Parameter

Any

**extract**

**Kind:** Method

```
def extract(self, subgraph: Subgraph, split_edge_fn: Any, internal_data: Any,
split_node: Any, owned_node: Any) -> DotGraph:
```

Extract a subgraph with Python callbacks to transform edge/node data.

**subgraph**

**Kind:** Parameter

Subgraph

**split\_edge\_fn**

**Kind:** Parameter

Any

**internal\_data**

**Kind:** Parameter

Any

**split\_node**

**Kind:** Parameter

Any

**owned\_node**

**Kind:** Parameter

Any

**\_\_getitem\_\_**

**Kind:** Method

**\_\_getitem\_\_**

**Kind:** Overload

```
def __getitem__(self, key: Hedge) -> DotHedgeData:
```

**key**

**Kind:** Parameter

Hedge

**\_\_getitem\_\_**

**Kind:** Overload

```
def __getitem__(self, key: NodeIndex) -> DotVertexData:
```

**key**

**Kind:** Parameter

NodeIndex

**\_\_getitem\_\_**

**Kind:** Overload

```
def __getitem__(self, key: EdgeIndex) -> DotEdgeData:
```

**key**

**Kind:** Parameter

EdgeIndex

Exhaustive reference · Source

## **DotGraphBuilder**

Builder for constructing DOT graphs programmatically.

Builder for constructing DOT graphs programmatically.

```
class DotGraphBuilder:
```

**Requires features:** extension-module, abi3-py310

**\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls) -> DotGraphBuilder:
```

Create a new, empty graph builder.

**add\_node**

**Kind:** Method

```
def add_node(self, data: Optional[DotVertexData] = None) -> NodeIndex:
```

Add a node and return its index.

**data**

**Kind:** Parameter

Optional[DotVertexData]

**Default:** None

**add\_edge**

**Kind:** Method

```
def add_edge(self, source: NodeIndex, sink: NodeIndex, data: Optional[DotEdgeData] =
None, orientation: Optional[Orientation] = None, source_hedge: Optional[DotHedgeData]
= None, sink_hedge: Optional[DotHedgeData] = None) -> None:
```

Add an edge between two nodes.

**source**

**Kind:** Parameter

NodeIndex

**sink**

**Kind:** Parameter

NodeIndex

**data**

**Kind:** Parameter

Optional[DotEdgeData]

**Default:** None

**orientation**

**Kind:** Parameter

Optional[Orientation]

**Default:** None

**source\_hedge**

**Kind:** Parameter

Optional[DotHedgeData]

**Default:** None

**sink\_hedge**

**Kind:** Parameter

Optional[DotHedgeData]

**Default:** None

**add\_external\_edge**

**Kind:** Method

```
def add_external_edge(self, source: NodeIndex, data: Optional[DotEdgeData] = None,
orientation: Optional[Orientation] = None, flow: Optional[Flow] = None, hedge:
Optional[DotHedgeData] = None) -> None:
```

Add a dangling (external) edge incident to a node.

**source**

**Kind:** Parameter

NodeIndex

**data**

**Kind:** Parameter

Optional[DotEdgeData]

**Default:** None

### **orientation**

**Kind:** Parameter

Optional[Orientation]

**Default:** None

### **flow**

**Kind:** Parameter

Optional[Flow]

**Default:** None

### **hedge**

**Kind:** Parameter

Optional[DotHedgeData]

**Default:** None

### **build**

**Kind:** Method

```
def build(self) -> DotGraph:
```

Build the graph and reset the builder.

Exhaustive reference · Source

### **DotHedgeData**

DOT hedge (half-edge) data.

DOT hedge (half-edge) data.

```
class DotHedgeData:
```

**Requires features:** extension-module, abi3-py310

### **statement**

**Kind:** Getter

```
def statement(self) -> Optional[str]:
```

Optional statement string.

### **id**

**Kind:** Getter

```
def id(self) -> Optional[Hedge]:
```

Optional hedge id.

### **port\_label**

**Kind:** Getter

```
def port_label(self) -> Optional[str]:
```

Optional port label.

### **compasspt**

**Kind:** Getter

```
def compasspt(self) -> Optional[str]:
```

Optional compass point as a string.

**`__new__`**

**Kind:** AssociatedFunction

```
def __new__(cls, statement: Optional[str] = None, id: Optional[int] = None,
port_label: Optional[str] = None, compasspt: Optional[str] = None) -> DotHedgeData:
```

Create hedge data from fields.

**`statement`**

**Kind:** Parameter

Optional[str]

**Default:** None

**`id`**

**Kind:** Parameter

Optional[int]

**Default:** None

**`port_label`**

**Kind:** Parameter

Optional[str]

**Default:** None

**`compasspt`**

**Kind:** Parameter

Optional[str]

**Default:** None

**`__repr__`**

**Kind:** Method

```
def __repr__(self) -> str:
```

Debug-style representation.

Exhaustive reference · Source

**`DotVertexData`**

DOT vertex data (name, optional index, and attribute statements).

DOT vertex data (name, optional index, and attribute statements).

class DotVertexData:

**Requires features:** extension-module, abi3-py310

**`name`**

**Kind:** Getter

```
def name(self) -> Optional[str]:
```

Optional vertex name.

**index****Kind:** Getter

```
def index(self) -> Optional[NodeIndex]:
```

Optional node index.

**statements****Kind:** Getter

```
def statements(self) -> NodeStatements:
```

Attribute statements as a dict-like proxy.

**\_\_new\_\_****Kind:** AssociatedFunction

```
def __new__(cls, name: Optional[str] = None, index: Optional[int] = None, statements: Optional[Mapping[str, str]] = None) -> DotVertexData:
```

Create vertex data from fields.

**name****Kind:** Parameter

Optional[str]

**Default:** None**index****Kind:** Parameter

Optional[int]

**Default:** None**statements****Kind:** Parameter

Optional[Mapping[str, str]]

**Default:** None**add\_statement****Kind:** Method

```
def add_statement(self, key: str, value: str) -> None:
```

Insert or update an attribute statement.

**key****Kind:** Parameter

str

**value****Kind:** Parameter

str

**\_\_repr\_\_****Kind:** Method

```
def __repr__(self) -> str:
Debug-style representation.
Exhaustive reference · Source
```

### **EdgeData**

Edge data with orientation.  
Edge data with orientation.  
class EdgeData:

**Requires features:** extension-module, abi3-py310

#### **orientation**

**Kind:** Getter

```
def orientation(self) -> Orientation:
Orientation of this edge.
```

#### **data**

**Kind:** Getter

```
def data(self) -> DotEdgeData:
DOT edge data.
```

#### **\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, orientation: Any, data: Any) -> EdgeData:
Create edge data from orientation and DotEdgeData.
```

#### **orientation**

**Kind:** Parameter

Any

#### **data**

**Kind:** Parameter

Any

#### **\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> str:
Debug-style representation.
Exhaustive reference · Source
```

### **EdgeIndex**

Edge index identifier.  
Edge index identifier.  
class EdgeIndex:

**Requires features:** extension-module, abi3-py310

**value**

**Kind:** Getter

```
def value(self) -> int:
```

Numeric value of this edge index.

**\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, value: int) -> EdgeIndex:
```

Create an edge index from a zero-based index.

**value**

**Kind:** Parameter

int

**\_\_int\_\_**

**Kind:** Method

```
def __int__(self) -> int:
```

Convert to int.

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

**EdgeStatements**

Dict-like proxy for edge attribute statements.

Dict-like proxy for edge attribute statements.

```
class EdgeStatements:
```

**Requires features:** extension-module, abi3-py310

**\_\_getitem\_\_**

**Kind:** Method

```
def __getitem__(self, key: str) -> str:
```

**key**

**Kind:** Parameter

str

**\_\_setitem\_\_**

**Kind:** Method

```
def __setitem__(self, key: str, value: str) -> None:
```

**key**

**Kind:** Parameter

str

**value**

**Kind:** Parameter

str

**\_\_delitem\_\_**

**Kind:** Method

def \_\_delitem\_\_(self, key: str) -> None:

**key**

**Kind:** Parameter

str

**\_\_contains\_\_**

**Kind:** Method

def \_\_contains\_\_(self, key: str) -> bool:

**key**

**Kind:** Parameter

str

**\_\_len\_\_**

**Kind:** Method

def \_\_len\_\_(self) -> int:

**\_\_iter\_\_**

**Kind:** Method

def \_\_iter\_\_(self) -> Iterator[str]:

**get**

**Kind:** Method

def get(self, key: str, default: Any = None) -> Any:

**key**

**Kind:** Parameter

str

**default**

**Kind:** Parameter

Any

**Default:** None

**keys**

**Kind:** Method

def keys(self) -> list[str]:

**values**

**Kind:** Method

def values(self) -> list[str]:

**items****Kind:** Method

```
def items(self) -> list[tuple[str, str]]:
```

**clear****Kind:** Method

```
def clear(self) -> None:
```

**update****Kind:** Method

```
def update(self, other: Mapping[str, str]) -> None:
```

**other****Kind:** Parameter

```
Mapping[str, str]
```

**pop****Kind:** Method

```
def pop(self, key: str, default: Any = None) -> Any:
```

**key****Kind:** Parameter

```
str
```

**default****Kind:** Parameter

```
Any
```

**Default:** None**popitem****Kind:** Method

```
def popitem(self) -> tuple[str, str]:
```

**setdefault****Kind:** Method

```
def setdefault(self, key: str, default: Optional[str] = None) -> str:
```

**key****Kind:** Parameter

```
str
```

**default****Kind:** Parameter

```
Optional[str]
```

**Default:** None**\_\_repr\_\_****Kind:** Method

```
def __repr__(self) -> str:
```

Exhaustive reference · Source

### **Flow**

Flow direction (Source/Sink).

Flow direction (Source/Sink).

```
class Flow:
```

**Requires features:** extension-module, abi3-py310

### **source**

**Kind:** AssociatedFunction

```
def source() -> Flow:
```

Flow::Source.

### **sink**

**Kind:** AssociatedFunction

```
def sink() -> Flow:
```

Flow::Sink.

### **\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> str:
```

Debug-style representation.

Exhaustive reference · Source

### **GlobalData**

Global graph attributes (graph/node/edge statements).

Global graph attributes (graph/node/edge statements).

```
class GlobalData:
```

**Requires features:** extension-module, abi3-py310

### **name**

**Kind:** Getter

```
def name(self) -> str:
```

Graph name.

### **name**

**Kind:** Setter

```
def name(self, value: str) -> None:
```

### **value**

**Kind:** Parameter

str

### **statements**

**Kind:** Getter

```
def statements(self) -> GlobalStatements:
```

Graph-level statements.

**statements**

**Kind:** Setter

```
def statements(self, value: dict[str, str]) -> None:
```

**value**

**Kind:** Parameter

dict[str, str]

**edge\_statements**

**Kind:** Getter

```
def edge_statements(self) -> GlobalStatements:
```

Default edge statements.

**edge\_statements**

**Kind:** Setter

```
def edge_statements(self, value: dict[str, str]) -> None:
```

**value**

**Kind:** Parameter

dict[str, str]

**node\_statements**

**Kind:** Getter

```
def node_statements(self) -> GlobalStatements:
```

Default node statements.

**node\_statements**

**Kind:** Setter

```
def node_statements(self, value: dict[str, str]) -> None:
```

**value**

**Kind:** Parameter

dict[str, str]

**\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, name: Optional[str] = None, statements: Optional[Mapping[str, str]]
= None, edge_statements: Optional[Mapping[str, str]] = None, node_statements:
Optional[Mapping[str, str]] = None) -> GlobalData:
```

Create global data from fields.

**name**

**Kind:** Parameter

Optional[str]

**Default:** None

**statements****Kind:** Parameter

Optional[Mapping[str, str]]

**Default:** None**edge\_statements****Kind:** Parameter

Optional[Mapping[str, str]]

**Default:** None**node\_statements****Kind:** Parameter

Optional[Mapping[str, str]]

**Default:** None**add\_statement****Kind:** Method

```
def add_statement(self, key: str, value: str) -> None:
```

Insert or update a graph-level statement.

**key****Kind:** Parameter

str

**value****Kind:** Parameter

str

**add\_edge\_statement****Kind:** Method

```
def add_edge_statement(self, key: str, value: str) -> None:
```

Insert or update a default edge statement.

**key****Kind:** Parameter

str

**value****Kind:** Parameter

str

**add\_node\_statement****Kind:** Method

```
def add_node_statement(self, key: str, value: str) -> None:
```

Insert or update a default node statement.

**key**

**Kind:** Parameter

str

**value**

**Kind:** Parameter

str

**\_\_repr\_\_**

**Kind:** Method

def \_\_repr\_\_(self) -> str:

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

**GlobalStatements**

Dict-like proxy for global statements.

Dict-like proxy for global statements.

class GlobalStatements:

**Requires features:** extension-module, abi3-py310

**\_\_getitem\_\_**

**Kind:** Method

def \_\_getitem\_\_(self, key: str) -> str:

**key**

**Kind:** Parameter

str

**\_\_setitem\_\_**

**Kind:** Method

def \_\_setitem\_\_(self, key: str, value: str) -> None:

**key**

**Kind:** Parameter

str

**value**

**Kind:** Parameter

str

**\_\_delitem\_\_**

**Kind:** Method

def \_\_delitem\_\_(self, key: str) -> None:

**key**

**Kind:** Parameter

str

**`__contains__`**

**Kind:** Method

```
def __contains__(self, key: str) -> bool:
```

**key**

**Kind:** Parameter

str

**`__len__`**

**Kind:** Method

```
def __len__(self) -> int:
```

**`__iter__`**

**Kind:** Method

```
def __iter__(self) -> Iterator[str]:
```

**get**

**Kind:** Method

```
def get(self, key: str, default: Any = None) -> Any:
```

**key**

**Kind:** Parameter

str

**default**

**Kind:** Parameter

Any

**Default:** None

**keys**

**Kind:** Method

```
def keys(self) -> list[str]:
```

**values**

**Kind:** Method

```
def values(self) -> list[str]:
```

**items**

**Kind:** Method

```
def items(self) -> list[tuple[str, str]]:
```

**clear**

**Kind:** Method

```
def clear(self) -> None:
```

**update**

**Kind:** Method

```
def update(self, other: Mapping[str, str]) -> None:
```

## **other**

**Kind:** Parameter

Mapping[str, str]

## **pop**

**Kind:** Method

def pop(self, key: str, default: Any = None) -> Any:

## **key**

**Kind:** Parameter

str

## **default**

**Kind:** Parameter

Any

**Default:** None

## **popitem**

**Kind:** Method

def popitem(self) -> tuple[str, str]:

## **setdefault**

**Kind:** Method

def setdefault(self, key: str, default: Optional[str] = None) -> str:

## **key**

**Kind:** Parameter

str

## **default**

**Kind:** Parameter

Optional[str]

**Default:** None

## **\_\_repr\_\_**

**Kind:** Method

def \_\_repr\_\_(self) -> str:

Exhaustive reference · Source

## **Hedge**

Half-edge identifier.

Half-edge identifier.

class Hedge:

**Requires features:** extension-module, abi3-py310

## **value**

**Kind:** Getter

```
def value(self) -> int:
 Numeric value of this hedge.
```

**\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, value: int) -> Hedge:
 Create a hedge from a zero-based index.
```

**value**

**Kind:** Parameter

int

**\_\_int\_\_**

**Kind:** Method

```
def __int__(self) -> int:
 Convert to int.
```

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> str:
 Debug-style representation.
```

Exhaustive reference · Source

**HedgeNode**

Hedge node with internal graph and hairs.

Hedge node with internal graph and hairs.

```
class HedgeNode:
```

**Requires features:** extension-module, abi3-py310

**internal\_graph**

**Kind:** Getter

```
def internal_graph(self) -> Subgraph:
 Internal subgraph.
```

**hairs**

**Kind:** Getter

```
def hairs(self) -> Subgraph:
 Hair subgraph.
```

**\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, internal_graph: Subgraph, hairs: Subgraph) -> HedgeNode:
 Create a hedge node from internal graph and hairs subgraphs.
```

**internal\_graph**

**Kind:** Parameter

Subgraph

**hairs**

**Kind:** Parameter

Subgraph

**\_\_repr\_\_**

**Kind:** Method

def \_\_repr\_\_(self) -> str:

Debug-style representation.

Exhaustive reference · Source

**HedgePair**

Hedge pairing (paired, unpaired, split).

Hedge pairing (paired, unpaired, split).

class HedgePair:

**Requires features:** extension-module, abi3-py310

**kind**

**Kind:** Getter

def kind(self) -> str:

Kind of hedge pair: \"paired\", \"unpaired\", or \"split\".

**source**

**Kind:** Getter

def source(self) -> Optional[Hedge]:

Source hedge (paired/split).

**sink**

**Kind:** Getter

def sink(self) -> Optional[Hedge]:

Sink hedge (paired/split).

**hedge**

**Kind:** Getter

def hedge(self) -> Optional[Hedge]:

Hedge (unpaired only).

**flow**

**Kind:** Getter

def flow(self) -> Optional[Flow]:

Flow for unpaired hedge.

**split**

**Kind:** Getter

def split(self) -> Optional[Flow]:

Which side is split for split edges.

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

**NodeIndex**

Node index identifier.

Node index identifier.

```
class NodeIndex:
```

**Requires features:** extension-module, abi3-py310

**value**

**Kind:** Getter

```
def value(self) -> int:
```

Numeric value of this node index.

**\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, value: int) -> NodeIndex:
```

Create a node index from a zero-based index.

**value**

**Kind:** Parameter

int

**\_\_int\_\_**

**Kind:** Method

```
def __int__(self) -> int:
```

Convert to int.

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

**NodeStatements**

Dict-like proxy for node attribute statements.

Dict-like proxy for node attribute statements.

```
class NodeStatements:
```

**Requires features:** extension-module, abi3-py310

**`__getitem__`**

**Kind:** Method

```
def __getitem__(self, key: str) -> str:
```

**key**

**Kind:** Parameter

str

**`__setitem__`**

**Kind:** Method

```
def __setitem__(self, key: str, value: str) -> None:
```

**key**

**Kind:** Parameter

str

**value**

**Kind:** Parameter

str

**`__delitem__`**

**Kind:** Method

```
def __delitem__(self, key: str) -> None:
```

**key**

**Kind:** Parameter

str

**`__contains__`**

**Kind:** Method

```
def __contains__(self, key: str) -> bool:
```

**key**

**Kind:** Parameter

str

**`__len__`**

**Kind:** Method

```
def __len__(self) -> int:
```

**`__iter__`**

**Kind:** Method

```
def __iter__(self) -> Iterator[str]:
```

**get**

**Kind:** Method

```
def get(self, key: str, default: Any = None) -> Any:
```

**key****Kind:** Parameter

str

**default****Kind:** Parameter

Any

**Default:** None**keys****Kind:** Method

def keys(self) -&gt; list[str]:

**values****Kind:** Method

def values(self) -&gt; list[str]:

**items****Kind:** Method

def items(self) -&gt; list[tuple[str, str]]:

**clear****Kind:** Method

def clear(self) -&gt; None:

**update****Kind:** Method

def update(self, other: Mapping[str, str]) -&gt; None:

**other****Kind:** Parameter

Mapping[str, str]

**pop****Kind:** Method

def pop(self, key: str, default: Any = None) -&gt; Any:

**key****Kind:** Parameter

str

**default****Kind:** Parameter

Any

**Default:** None**popitem****Kind:** Method

```
def popitem(self) -> tuple[str, str]:
```

**setdefault**

**Kind:** Method

```
def setdefault(self, key: str, default: Optional[str] = None) -> str:
```

**key**

**Kind:** Parameter

str

**default**

**Kind:** Parameter

Optional[str]

**Default:** None

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> str:
```

Exhaustive reference · Source

**Orientation**

Edge orientation (Default/Reversed/Undirected).

Edge orientation (Default/Reversed/Undirected).

class Orientation:

**Requires features:** extension-module, abi3-py310

**default**

**Kind:** AssociatedFunction

```
def default() -> Orientation:
```

Orientation::Default.

**reversed**

**Kind:** AssociatedFunction

```
def reversed() -> Orientation:
```

Orientation::Reversed.

**undirected**

**Kind:** AssociatedFunction

```
def undirected() -> Orientation:
```

Orientation::Undirected.

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> str:
```

Debug-style representation.

Exhaustive reference · Source

### **OrientedCut**

Oriented cut represented by left/right subgraphs.

Oriented cut represented by left/right subgraphs.

class OrientedCut:

**Requires features:** extension-module, abi3-py310

#### **left**

**Kind:** Getter

def left(self) -> Subgraph:

Left side of the cut.

#### **right**

**Kind:** Getter

def right(self) -> Subgraph:

Right side of the cut.

#### **\_\_repr\_\_**

**Kind:** Method

def \_\_repr\_\_(self) -> str:

Debug-style representation.

Exhaustive reference · Source

### **Subgraph**

Subgraph represented as a bitset of hedges.

Subgraph represented as a bitset of hedges.

class Subgraph:

**Requires features:** extension-module, abi3-py310

#### **empty**

**Kind:** AssociatedFunction

def empty(cls, size: int) -> Subgraph:

Create an empty subgraph of the given size (number of hedges).

#### **size**

**Kind:** Parameter

int

#### **full**

**Kind:** AssociatedFunction

def full(cls, size: int) -> Subgraph:

Create a full subgraph including all hedges.

#### **size**

**Kind:** Parameter

int

**from\_hedges****Kind:** AssociatedFunction

```
def from_hedges(cls, size: int, hedges: Sequence[Hedge]) -> Subgraph:
```

Create a subgraph from a list of hedges.

**size****Kind:** Parameter

int

**hedges****Kind:** Parameter

Sequence[Hedge]

**to\_hedges****Kind:** Method

```
def to_hedges(self) -> list[Hedge]:
```

List included hedges.

**size****Kind:** Method

```
def size(self) -> int:
```

Total number of hedges in the parent graph.

**n\_included****Kind:** Method

```
def n_included(self) -> int:
```

Number of included hedges.

**includes****Kind:** Method

```
def includes(self, hedge: Any) -> bool:
```

Whether a hedge is included.

**hedge****Kind:** Parameter

Any

**union****Kind:** Method

```
def union(self, other: Subgraph) -> Subgraph:
```

Union with another subgraph.

**other****Kind:** Parameter

Subgraph

## **intersection**

**Kind:** Method

```
def intersection(self, other: Subgraph) -> Subgraph:
Intersection with another subgraph.
```

## **other**

**Kind:** Parameter

Subgraph

## **sym\_diff**

**Kind:** Method

```
def sym_diff(self, other: Subgraph) -> Subgraph:
Symmetric difference with another subgraph.
```

## **other**

**Kind:** Parameter

Subgraph

## **subtract**

**Kind:** Method

```
def subtract(self, other: Subgraph) -> Subgraph:
Subtract another subgraph.
```

## **other**

**Kind:** Parameter

Subgraph

## **\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> str:
Debug-style representation.

Exhaustive reference · Source
```

## **TraversalTree**

Traversal tree produced by DFS/BFS.

Traversal tree produced by DFS/BFS.

```
class TraversalTree:
```

**Requires features:** extension-module, abi3-py310

## **tree\_subgraph**

**Kind:** Method

```
def tree_subgraph(self) -> Subgraph:
Subgraph corresponding to the tree edges.
```

## **node\_order**

**Kind:** Method

```
def node_order(self) -> list[NodeIndex]:
```

Node order from traversal.

### **covers**

**Kind:** Method

```
def covers(self, subgraph: Subgraph) -> Subgraph:
```

Covers a subgraph with the traversal tree.

### **subgraph**

**Kind:** Parameter

Subgraph

### **iter\_hedges**

**Kind:** Method

```
def iter_hedges(self) -> list[tuple[Hedge, str, Optional[Hedge]]]:
```

Iterate hedges as (hedge, kind, root\_hedge).

Exhaustive reference · Source

## Version information

Save these identifiers with published results and include them in issue reports so that collaborators can reproduce the same software environment.

- **Documentation channel:** latest
- **Documentation route:** /products/linnet/latest/
- **Source revision:** cbfc3a5827679de14e9373edd6b332e6b9b28a55

## Package versions and enabled features

- gammaloop/gammalooprs — 0.3.4 (no optional features)
- gammaloop/gammaloop-api — 0.1.0 (features: cli)
- gammaloop/gammaloop — 0.3.4 (features: python\_api, python\_abi, python\_stubgen)
- linnet/linnet — 0.17.0 (features: nodestore-vec, serde, bincode, drawing, symbolica)
- linnet/linnet-py — 0.1.0 (features: extension-module, abi3-py310)
- spenso/spenso — 0.6.0 (features: shadowing)
- spenso/spenso-macros — 0.3.2 (features: shadowing)
- spenso/spenso-hep-lib — 0.1.7 (no optional features)
- spenso/spynso3 — 0.1.2 (features: python\_stubgen)
- idenso/idenso — 0.3.0 (features: bincode, reference-cases)
- idenso/idenso — 0.3.0 (features: python, python\_stubgen)
- vakint/vakint — 0.1.2 (no optional features)
- vakint/vakint — 0.1.2 (features: symbolica\_community\_module, python\_stubgen)